

Helix VPN — MVP2 Implementation Roadmap

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Revision 2 changelog: Hardening pass — added per-phase Test Plan cross-references (citing MVP2_SECURITY_PERFORMANCE.md §8 Testing Strategy and MVP2_OVERVIEW.md §9 SC-xx success criteria IDs) to every phase's Exit Criteria; added the four previously-absent cross-cutting production-readiness milestones (staged/canary rollout rehearsal, MDM/managed-policy validation, consent-gated crash/telemetry pipeline, reproducible-build + SBOM generation) with owners and exit criteria across Phase 1 (foundational tooling) and Phase 9 (validation + go-live execution); added `apply_managed_policy()` to the PlatformAdapter trait and per-platform managed-policy hook tasks in Phases 3/4/5/6/8; reconciled the connection-lifecycle state machine in Phase 1 Week 2 to the canonical MVP2_ARCHITECTURE.md §5.6 state set; corrected two "build a design system from scratch" tasks (Phase 3 Week 11, Phase 5 Week 14) to instead consume the existing OpenDesign token system; added a Mermaid gantt phase timeline to §1. No phase week-numbers or §13 Risk-Adjusted Timeline figures were changed — those remain the corpus-wide anchor.

Cross-Platform Development Plan: 36-Week Phased Delivery

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Derived From: MVP2_ARCHITECTURE.md, MVP2_SHARED_CORE.md

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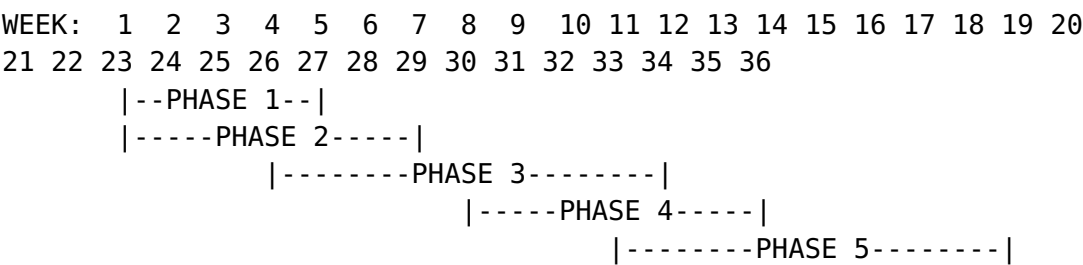
1. Roadmap Overview

1.1 Executive Summary

This roadmap translates the MVP2 architecture and shared core specifications into a concrete, phased execution plan spanning **36 weeks (9 months)** of active development. The plan delivers Helix VPN across **eight target platforms** organized into four priority tiers, with explicit dependency management, resource allocation, and exit criteria for each phase.

The roadmap is designed with **strategic phase overlap** to maximize parallel development and minimize calendar time. The shared Rust core (helix-core) serves as the critical path foundation — all other phases branch from its completion.

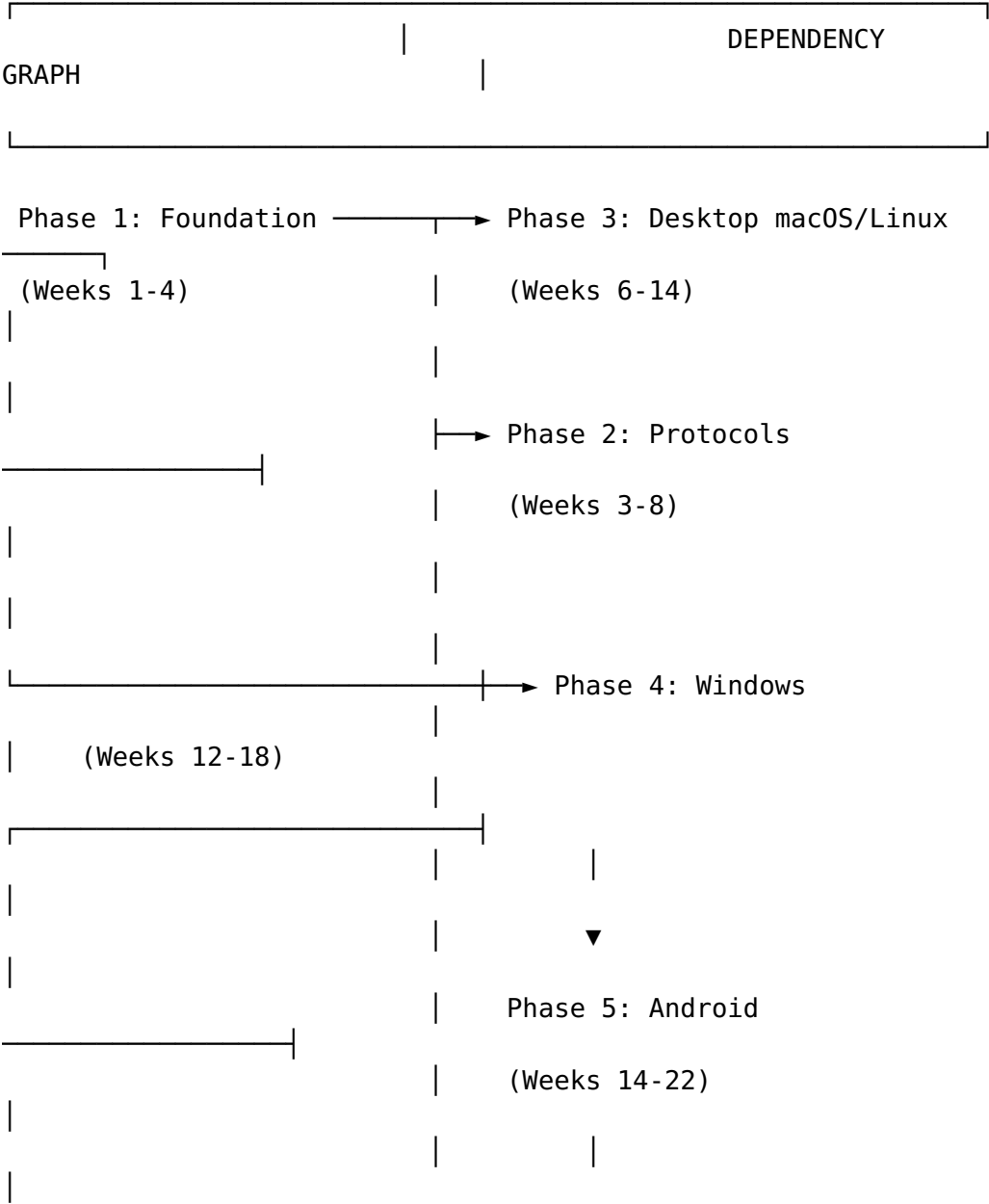
1.2 Total Timeline Summary

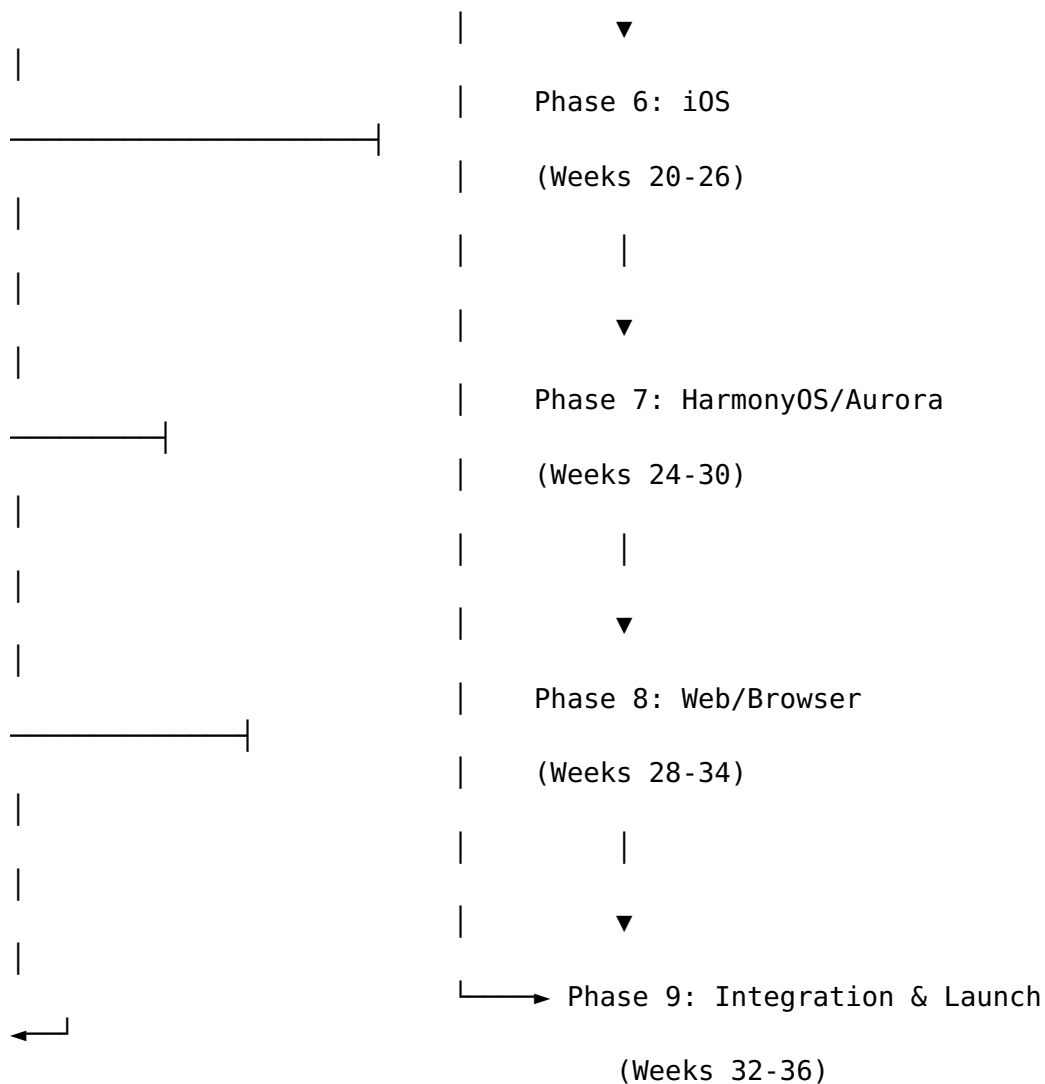




Total Duration: 36 weeks
Parallel Workstreams: 2-3 at peak (Weeks 14-28)
Critical Path: Phase 1 → Phase 2 → Phase 3 → Phase 5 → Phase 6 → Phase 9
Buffer Weeks: 2 weeks (Weeks 35-36 serve as contingency)

1.3 Phase Dependency Graph





Key Dependency Rules:

- Phases 3-8 all depend on Phase 1 (core library) completing at least the API surface
- Phase 2 (protocols) must complete WireGuard before Phase 3 desktop integration testing
- Phase 4 (Windows) can begin UI work before Phase 3 completes, but needs Phase 3 UI patterns
- Phase 5 (Android) and Phase 6 (iOS) share the Flutter mobile codebase — iOS trails Android by 6 weeks
- Phase 7 (HarmonyOS/Aurora) leverages mobile patterns from Phase 5-6
- Phase 8 (Web) has the loosest coupling — primarily depends on core crypto WASM builds
- Phase 9 (Integration) requires at least one client from each tier to be feature-complete

1.4 Platform Delivery Order

Priority Tier	Platform	Target Week	Entry Criteria
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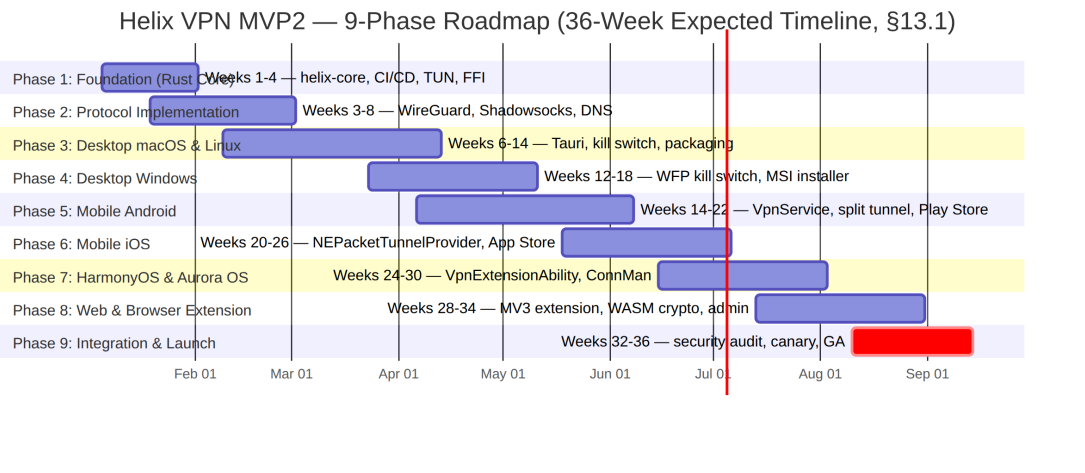
Tier 1 (P0) milestone	macOS	Week 10	Desktop Alpha
Tier 1 (P0) milestone	Linux	Week 12	Desktop Alpha
Tier 1 (P0) milestone	Windows	Week 16	Desktop Beta
Tier 1 (P0) milestone	Android	Week 20	Mobile Alpha
Tier 2 (P1) milestone	iOS	Week 24	Mobile Beta
Tier 2 (P1) Platforms milestone	HarmonyOS	Week 28	Extended
Tier 3 (P2) Platforms milestone	Aurora OS	Week 28	Extended
Tier 4 (P3) milestone	Web/Browser	Week 32	Web Complete
All Platforms milestone	Production	Week 36	Launch Ready

1.5 Code Reuse Targets by Phase

Phase	Code Being Written	Estimated Reuse %	Lines of Code (est.)
Phase 1	Core library, abstractions	100% (foundation)	8,000
Phase 2	WireGuard, Shadowsocks, DNS	100% (foundation)	6,000
Phase 3	macOS/Linux desktop UI + adapters	85% core reuse	4,000
Phase 4	Windows desktop UI + adapters	80% core reuse	3,500
Phase 5	Android mobile UI + adapters	72% core reuse	5,000
Phase 6	iOS mobile UI + adapters	72% core reuse	3,000
Phase 7	HarmonyOS/Aurora adaptations	70-75% core reuse	4,500
Phase 8	Browser extension, WASM, admin	45% core reuse	5,000
Phase 9	Integration, testing, docs	N/A	3,000
TOTAL		~78% weighted avg	~42,000

1.6 Phase Timeline (Mermaid Gantt)

The ASCII summary in §1.2 and the dependency graph in §1.3 are the normative week-range source; the chart below is a rendering of the same 9 phases against the 36-week **expected** timeline (§13.1) for tooling that renders Mermaid. Dates are illustrative placeholders anchored to a nominal "Week 1 = Monday" project start — only the **week ranges** (identical to each phase's own heading) are authoritative, not the calendar dates themselves, which are fixed at project kickoff.



2. Phase 1: Foundation — Rust Core Library (Weeks 1-4)

2.1 Phase Overview

Attribute	Value
Duration	4 weeks
Team	2 Rust Core Engineers + 1 DevOps
Depends On	None (first phase)
Blocks	All subsequent phases
Goal	Production-grade Rust core library with CI/CD, basic TUN, and extensible abstractions

2.2 Week-by-Week Breakdown

Week 1: Project Structure & CI/CD Pipeline

Tasks:



Initialize Git repository with branch protection rules (main, develop, release/*)



Create Cargo workspace structure per architecture spec:

```
helix-core/
├── Cargo.toml                    # Workspace manifest
├── crates/
│   ├── helix-core-api/         # Public FFI API surface
│   ├── helix-vpn-engine/       # Generic tunnel management
│   ├── helix-wireguard/        # WireGuard protocol
│   └── (placeholder)
│       ├── helix-crypto/       # Encryption primitives
wrapper
├── helix-network/              # HTTP client, latency
testing
├── helix-platform-abstraction/ # OS adapter trait
definitions
├── bindings/
│   ├── uniffi/                 # UniFFI UDL definitions
│   └── wasm/                   # WASM target config
├── tests/
│   └── integration/            # Cross-platform integration
tests
├── fuzz/                       # Crypto fuzzing targets
└── .github/workflows/         # CI/CD pipeline definitions
```



Set up GitHub Actions CI pipeline with matrix builds:

- Linux x64 (Tier 1) — ubuntu-latest
- Linux ARM64 (Tier 1) — ubuntu-latest + cross
- macOS x64 + ARM64 (Tier 2) — macos-latest
- Windows x64 (Tier 1) — windows-latest
- WASM (Tier 2) — ubuntu-latest + wasm-pack
- CI stages: fmt → clippy → test → build → doc



Configure cross-compilation toolchain (cross-rs, cargo-ndk)



Set up dependabot for automated dependency updates



Configure cargo-deny for license and security auditing



Set up code coverage with tarpaulin + Codecov integration



Define rustfmt.toml and enforce formatting in CI



Production-readiness foundation — reproducible builds + SBOM (lands here, not bolted on post-launch, because every subsequent phase's CI job inherits this pipeline stage; cross-ref

MVP2_SHARED_CORE.md §5.5 Supply-Chain & Build Integrity + MVP2_ARCHITECTURE.md §10.3):

- Pin toolchain via rust-toolchain.toml; build with `cargo build --locked` against a committed `Cargo.lock` so every artifact is bit-for-bit reproducible from the same commit
- Add cargo-cyclonedx (or syft) as a CI stage generating a CycloneDX SBOM for every cross-compiled artifact, published as a build artifact
- Owner: DevOps/CI Engineer

Deliverables:

- Repository initialized with full CI/CD pipeline
- Matrix builds passing for Tier 1 targets (Linux x64, Windows x64)
- Developer onboarding documentation (CONTRIBUTING.md, BUILD.md)
- Reproducible-build configuration (pinned toolchain + locked lockfile) and a working SBOM-generation CI stage, both proven against a sample build

Exit Criteria:

- CI pipeline green on all Tier 1 targets
 - `cargo build` succeeds for workspace
 - `cargo test` passes with 100% compilation
 - A sample CI build produces an identical binary hash on two independent runs (reproducibility proof) and an attached SBOM artifact (MVP2_SHARED_CORE.md §5.5); this is the foundation the Phase 9 Week 33/36 SBOM gate re-verifies against the actual release artifacts — see §10.2
-

Week 2: Core Abstractions & Trait Definitions

Tasks:



Implement TunnelDevice trait (TUN interface abstraction):

```
#[async_trait]
pub trait TunnelDevice: Send + Sync {
    async fn read_packet(&self, buf: &mut [u8]) -> Result<usize>;
    async fn write_packet(&self, packet: &[u8]) -> Result<()>;
    fn set_mtu(&self, mtu: u16) -> Result<()>;
    fn get_interface_name(&self) -> String;
}
```



Implement PlatformAdapter trait (OS-specific VPN lifecycle):


```
#[async_trait]
pub trait PlatformAdapter: Send + Sync {
    async fn setup_tunnel(&self, config: &TunnelConfig) ->
        Result<Box<dyn TunnelDevice>>;
    async fn protect_socket(&self, socket: RawFd) ->
        Result<()>;
    async fn configure_routes(&self, routes: &[Route]) ->
        Result<()>;
    async fn configure_dns(&self, servers: &[IpAddr]) ->
        Result<()>;
    async fn enable_kill_switch(&self) -> Result<()>;
    async fn disable_kill_switch(&self) -> Result<()>;
    async fn set_split_tunnel_apps(&self, allowed: &[String],
        blocked: &[String]) -> Result<()>;
    async fn get_default_interface(&self) ->
        Result<NetworkInterface>;
    async fn on_network_change(&self, callback:
        NetworkChangeCallback) -> Result<()>;
    async fn apply_managed_policy(&self, policy:
        &ManagedPolicy) -> Result<()>;
}
```

- `apply_managed_policy()` is the enterprise-fleet extension point: helix-admin pushes org-scoped policy (allowed protocols, forced kill switch, forced split-tunnel rules, DNS policy) via the MVP1 Admin API, and each platform's native MDM/managed-config channel invokes this method (cross-ref MVP2_ARCHITECTURE.md §10.2). Per-platform implementations land in Phases 3-6/8 (macOS Configuration Profile / Windows Group Policy+Intune / Android Enterprise AppConfig / Apple MDM managed app config / Chrome ExtensionSettings); end-to-end validation against real MDM channels happens in Phase 9 §10.2 Week 32



Implement `ProtocolDriver` trait (protocol-agnostic interface):

```
#[async_trait]
pub trait ProtocolDriver: Send + Sync {
    async fn handshake(&self, endpoint: SocketAddr, port: u16)
        -> Result<TunnelParams>;
    async fn start_io(&self, tun_tx: mpsc::Sender<Bytes>,
        net_rx: mpsc::Receiver<Bytes>, mtu: u16) ->
        Result<()>;
    async fn shutdown(&self) -> Result<()>;
}
```



Implement `Firewall` trait (kill switch abstraction):

```
#[async_trait]
pub trait Firewall: Send + Sync {
    async fn enable_kill_switch(&self) -> Result<()>;
    async fn disable_kill_switch(&self) -> Result<()>;
}
```

```

    async fn allow_endpoint(&self, address: SocketAddr) ->
        Result<()>;
    async fn block_ipv6(&self) -> Result<()>;
    async fn enforce_dns(&self, dns_servers: &[String]) ->
        Result<()>;
}

```



Define ConnectionStateMachine with validated transitions, matching the canonical state set in MVP2_ARCHITECTURE.md §5.6 (extended from an earlier, thinner draft to include reconnect and kill-switch-active states):

- Disconnected → Connecting → Connected → Reconnecting → KillSwitchActive → Disconnecting → Disconnected
- Plus the terminal/error state ConnectionFailed, reachable from Connecting, Connected, and Reconnecting
- ConnectionFailed → Disconnected | Connecting (retry)
- Reconnecting → Connected (recovery) or Reconnecting → KillSwitchActive (recovery exhausted, kill switch engages per configured policy)



Implement error hierarchy (HelixError enum with thiserror):

- ConfigParse, ConnectionFailed, AuthenticationFailed, Timeout, StateTransition, CryptoError, PlatformError, NetworkError



Set up structured logging with tracing + tracing-subscriber



Implement SecureStorage trait (keychain/keystore abstraction)

Deliverables:

- All core traits defined and documented with rustdoc
- Unit tests for state machine transition validation
- Mock implementations of all traits for testing

Exit Criteria:

- 100% trait compilation across all targets
- State machine unit tests passing
- Documentation coverage > 80%

Week 3: Basic TUN Interface (Linux & macOS)

Tasks:



Implement Linux TUN adapter using tun-rs crate:

- Create TUN device via /dev/net/tun ioctl
- Configure IP address and MTU via netlink
- Set blocking/async I/O modes

- Implement `read_packet()` / `write_packet()` with zero-copy where possible
- ☐ Implement macOS TUN adapter using `utun` kernel interface:
 - Create `utun` interface via `PF_SYSTEM` socket domain
 - Configure with `ifconfig` / `ioctl` calls
 - Handle macOS-specific `utun` header stripping
- ☐ Implement basic Linux routing via `rtnetlink`:
 - Add default route through TUN interface
 - Preserve existing routes for VPN server endpoint
 - Route restoration on disconnect
- ☐ Implement basic macOS routing via `route` command + `sysctl`:
 - Add route to VPN server via primary interface
 - Redirect default route to `utun` interface
- ☐ Create mock TUN device for unit testing (no root required)
- ☐ Write integration test: create TUN → write packet → read packet → destroy TUN

Deliverables:

- Working TUN implementation on Linux and macOS
- Routing configuration for full-tunnel mode
- Integration test suite for TUN lifecycle

Exit Criteria:

- Integration test passes on Linux (Ubuntu 22.04+)
 - Integration test passes on macOS (12+)
 - No memory leaks (verified with `valgrind` / `leaks`)
-

Week 4: FFI Layer & Public API Surface

Tasks:

- ☐ Implement C FFI layer (`src/platform/ffi.rs`):
 - `helix_init()` — initialize engine with JSON config
 - `helix_connect()` — establish VPN connection
 - `helix_disconnect()` — disconnect VPN
 - `helix_get_state()` — query connection state
 - `helix_free()` / `helix_free_string()` — memory management
 - `helix_last_error()` — error retrieval
 - Thread-safe static Tokio runtime with `LazyLock<Runtime>`
 - Panic catching with `catch_unwind`
- ☐

Set up UniFFI scaffolding:

- Create `helix_core.udl` interface definition
- Configure `uniffi.toml` for Kotlin and Swift output
- Generate initial bindings (manual verification)



Set up `flutter_rust_bridge` scaffolding:

- Create bridge module with `#[frb]` annotations
- Implement `init_app()` and basic async commands



Configure `cbindgen.toml` for C header generation (Aurora OS)



Implement `HelixVpnApi` public API struct:

- `new(config: VpnConfig) -> Result<Self>`
- `connect() -> Result<ConnectionStatus>`
- `disconnect() -> Result<ConnectionStatus>`
- `status() -> ConnectionStatus`
- `subscribe_events() -> broadcast::Receiver<EngineEvent>`
- `stats() -> ConnectionStats`



Write FFI integration tests:

- C FFI round-trip test (init → connect → disconnect → free)
- UniFFI binding compilation test
- FRB bridge smoke test

Deliverables:

- Complete FFI layer with C ABI stability
- UniFFI UDL definitions for Kotlin (Android) and Swift (iOS)
- FRB bridge module for Flutter integration
- API documentation for all public functions

Exit Criteria:

- C FFI test passes (memory-safe, no leaks)
- UniFFI bindings compile for Kotlin and Swift
- FRB bridge generates valid Dart bindings
- All public APIs documented

2.3 Phase 1 Deliverables Summary

Deliverable	Description	Owner
helix-core workspace	Full Cargo workspace with 6 crates	Rust Core Lead
CI/CD pipeline	GitHub Actions with matrix builds	DevOps
Core traits	TunnelDevice, PlatformAdapter, ProtocolDriver, Firewall	Rust Core Lead

Deliverable	Description	Owner
TUN implementation	Linux + macOS TUN with routing	Rust Core Engineer
FFI layer	C ABI, UniFFI, FRB scaffolding	Rust Core Engineer
Public API	HelixVpnApi with full documentation	Rust Core Lead
Test suite	Unit + integration tests, >70% coverage	Both
Supply-chain tooling	Pinned toolchain, cargo-cyclonedx/syft SBOM CI stage	DevOps

2.4 Phase 1 Exit Criteria

- ☐ All Tier 1 CI targets building successfully
- ☐ Core traits implemented and documented (incl. `apply_managed_policy()` on `PlatformAdapter` and the canonical `ConnectionStateMac` hine per `MVP2_ARCHITECTURE.md` §5.6)
- ☐ TUN interface working on Linux and macOS
- ☐ FFI layer complete with passing tests
- ☐ API documentation coverage > 80%
- ☐ Code review complete for all PRs
- ☐ No open security warnings from `cargo-audit`
- ☐ Reproducible build + SBOM generation wired into CI and proven against a sample artifact (foundation for the Phase 9 release-gate re-check)

Test Plan (cross-reference — do not re-derive a framework here): Unit tests per `MVP2_SECURITY_PERFORMANCE.md` §8.1 (Rust core unit coverage) + §8.7 (CI test matrix); relevant success criteria: SC-11 (code reuse), SC-21 (zero critical/high CVEs — `cargo-audit`), SC-22 (zero memory-safety warnings), SC-27 ($\geq 85\%$ core unit-test line coverage) per `MVP2_OVERVIEW.md` §9.

3. Phase 2: Protocol Implementation (Weeks 3-8)

3.1 Phase Overview

Attribute	Value
Duration	6 weeks (overlaps with Phase 1 from Week 3)
Team	2 Rust Core Engineers
Depends On	Phase 1 traits and abstractions (Week 2+)
Blocks	Phase 3 (desktop integration testing), Phase 5 (Android)
Goal	Production-grade WireGuard protocol, Shadowsocks, DNS resolver, and protocol negotiation

3.2 Week-by-Week Breakdown

Week 3 (Overlap): Crypto Primitives Foundation

Tasks:



Implement helix-crypto crate:

- X25519 keypair generation and shared secret derivation (x25519-dalek)
- ChaCha20-Poly1305 AEAD encryption/decryption (chacha20poly1305)
- Blake2s keyed hashing for WireGuard
- HKDF key derivation
- SecretKey type with ZeroizeOnDrop for automatic key wiping



Implement KeyManager for key lifecycle:

- Generate static keypairs
- Derive ephemeral keys for handshakes
- Session key rotation
- Secure key storage interface (delegates to platform)



Write crypto unit tests:

- X25519 key exchange correctness (known-answer tests)
- ChaCha20-Poly1305 encryption/decryption round-trip
- Blake2s test vectors
- Key zeroization verification



Set up crypto fuzzing targets with cargo-fuzz:

- ChaCha20-Poly1305 decryption fuzzing

- X25519 public key validation fuzzing

Deliverables:

- helix-crypto crate with full test coverage
 - Fuzzing targets configured and running
 - Crypto benchmark suite (baseline for performance regression)
-

Week 4 (Overlap): WireGuard Protocol — Handshake**Tasks:**

Integrate boringtun crate as WireGuard base:

- Fork or vendor boringtun for customization control
- Integrate boringtun::noise::Tunn for per-peer tunnel state
- Configure rate limiter for DoS protection



Implement Noise Protocol handshake (IK pattern):

- format_handshake_initiation() — create initiation message
- Handle handshake response with retry logic (5 retries, 5s interval)
- Derive session keys from handshake
- Cookie mechanism for IP obfuscation (anti-DoS)



Implement handshake state machine:

- Idle → InitiationSent → ResponseReceived → Established
- Timeout handling and retry logic
- Rekey after 120s or on demand



Write handshake tests:

- Successful handshake against test server
- Timeout and retry behavior
- Invalid response rejection
- Rekey trigger verification

Deliverables:

- WireGuard handshake implementation
 - Handshake integration tests
 - Performance benchmark: handshake < 500ms on reference hardware
-

Week 5: WireGuard Protocol — Transport & Timers**Tasks:**

Implement packet encapsulation/decapsulation:

- `encapsulate()` — encrypt IP packet with ChaCha20-Poly1305, prepend WG header
- `decapsulate()` — parse WG header, decrypt payload, validate MAC
- Handle IPv4 and IPv6 packets



Implement keepalive timer system:

- 25s persistent keepalive interval
- Rekey after time: 120s
- Rekey attempt time: 90s
- Rekey timeout: 5s



Implement packet counter and replay protection:

- 64-bit sender/receiver nonces
- Sliding window for replay detection
- Counter persistence across reconnects



Implement transport I/O loop:

- Async UDP socket send/receive via `tokio::net::UdpSocket`
- `bytes::Bytes` for zero-copy buffer management
- Channel-based communication with TUN layer



Write transport stress tests:

- 1000 packets/second sustained throughput
- Packet loss simulation and recovery
- Rekey during active traffic

Deliverables:

- Full WireGuard transport layer
- Timer system with passing tests
- Transport performance benchmarks

Week 6: Shadowsocks Protocol (SIP022 AEAD-2022)

Tasks:



Implement Shadowsocks protocol driver:

- SIP022 AEAD-2022 cipher support:
 - 2022-blake3-aes-256-gcm
 - 2022-blake3-chacha20-poly1305
- Session-based key derivation with BLAKE3
- Full replay protection with 64-bit unique session IDs
- TCP: length-chunk-payload-chunk model
- UDP: separate header encryption + AEAD body



Implement ShadowsocksDriver struct implementing ProtocolDriver:

r:

- handshake() — derive session keys, establish connection
- start_io() — relay encrypted traffic
- shutdown() — cleanup sessions



Write Shadowsocks tests:

- Connection to test Shadowsocks server
- Encryption/decryption correctness
- Session key rotation

Deliverables:

- Shadowsocks protocol driver
 - SIP022 AEAD-2022 cipher support
 - Integration tests against test server
-

Week 7: DNS Resolver (DoH/DoT)

Tasks:



Implement DnsResolver supporting multiple methods:

- Plaintext UDP/53 (fallback)
- DNS-over-TLS (DoT) via rustls
- DNS-over-HTTPS (DoH) via hickory-resolver
- DNS-over-HTTP/3 (DoH3) via QUIC (future)



Implement DNS caching layer:

- TTL-aware cache with eviction
- Prefetch for frequently queried domains
- Cache poisoning detection



Implement DNS leak prevention:

- Force all DNS through VPN tunnel
- Block non-VPN DNS queries when connected
- Per-platform DNS configuration



Write DNS resolver tests:

- Resolution correctness for all methods
- Cache hit/miss behavior
- Timeout and fallback behavior
- DNS leak prevention verification

Deliverables:

- Multi-method DNS resolver
 - DNS caching with leak prevention
 - DNS test suite
-

Week 8: Protocol Negotiation & Selection

Tasks:



Implement protocol negotiation system:

- Server capability advertisement (REST API)
- Client-side protocol preference ranking
- Automatic fallback on connection failure
- Latency-based protocol selection



Implement ProtocolSelector:

- `select_best_protocol(server_list, preferences) -> ProtocolConfig`
- Parallel latency testing across protocols
- Obfuscation auto-detection (censorship evasion)



Implement multi-hop chaining (MVP2 stretch goal):

- Entry-exit node separation
- Double encapsulation (WireGuard over WireGuard)
- Entry node routing configuration



Write protocol negotiation tests:

- Fallback behavior simulation
- Latency measurement accuracy
- Multi-hop connection chain

Deliverables:

- Protocol negotiation and selection system
- Multi-hop chaining support
- Protocol selection benchmarks

3.3 Phase 2 Deliverables Summary

Deliverable	Description	Test Coverage
helix-crypto	X25519, ChaCha20-Poly1305, Blake2s, HKDF	> 90%
helix-wireguard	Full WireGuard protocol (boringtun-based)	> 85%
helix-shadowsocks	SIP022 AEAD-2022 Shadowsocks	> 80%
helix-dns	DoH/DoT resolver with caching	> 85%
Protocol negotiation	Auto-selection with fallback	> 75%

Deliverable	Description	Test Coverage
Fuzzing corpus	Running 24/7 on CI	N/A

3.4 Phase 2 Exit Criteria

- ☐ WireGuard handshake completes in < 500ms (p99)
- ☐ WireGuard transport sustains 1000 packets/second
- ☐ Shadowsocks connects and encrypts correctly
- ☐ DNS resolver passes all method tests
- ☐ Protocol negotiation fallback works end-to-end
- ☐ All crypto fuzzing targets running without crashes
- ☐ No open security warnings from cargo-audit

Test Plan (cross-reference): Unit + property-based crypto tests per MVP2_SECURITY_PERFORMANCE.md §8.1; integration/connection tests per §8.2; performance tests (handshake latency, throughput) per §8.4; relevant success criteria: SC-02 (WireGuard handshake success rate, 1,000-connection stress test), SC-03 (Shadowsocks obfuscation vs. simulated DPI), SC-13 (connection time), SC-15 (throughput >= 500 Mbps desktop reference), SC-21/SC-22 (security/ memory-safety) per MVP2_OVERVIEW.md §9.

4. Phase 3: Desktop Clients — macOS & Linux (Weeks 6-14)

4.1 Phase Overview

Attribute	Value
Duration	9 weeks (overlaps with Phase 2 from Week 6)
Team	2 Desktop Developers + 1 UI/UX Designer
Depends On	Phase 1 (core API), Phase 2 WireGuard (Week 4+)
Blocks	Phase 4 (Windows builds on macOS/Linux patterns)
Goal	

Attribute	Value
	Fully functional macOS and Linux desktop clients with kill switch

4.2 Week-by-Week Breakdown

Week 6: Tauri v2 Desktop App Scaffold

Tasks:



Initialize Tauri v2 project:

```

helix-desktop/
├── src-tauri/                                # Rust backend
│   ├── Cargo.toml
│   ├── tauri.conf.json
│   └── src/
│       ├── main.rs                          # Entry point
│       └── commands.rs                      # Tauri commands (connect,
disconnect, etc.)
│   ├── state.rs                            # Application state management
│   └── core_bridge.rs                       # helix-core integration
├── capabilities/                           # Tauri capability definitions
├── src/                                    # React frontend
│   ├── components/                         # UI components
│   ├── pages/                             # Route pages
│   ├── store/                             # State management (Zustand)
│   └── styles/                             # CSS/Tailwind
└── package.json

```



Configure Tauri v2 with capability-based security:

- Define default capability with minimum permissions
- Add shell:allow-execute for route/firewall commands
- Scope permissions to specific binaries and paths



Implement core Tauri commands:

- vpn_connect(server_id, protocol)
- vpn_disconnect()
- get_connection_state()
- get_server_list()
- get_stats()



Set up frontend scaffold:

- React 19 + TypeScript + Vite
- Tailwind CSS for styling
- Zustand for state management
- React Query for server data fetching

Deliverables:

- Tauri app compiles and launches on macOS and Linux
 - Basic "Connect/Disconnect" button functional (mock core)
 - UI renders correctly in system WebView
-

Week 7: macOS NetworkExtension Integration

Tasks:



Create macOS Network Extension target:

- HelixPacketTunnel — NEPacketTunnelProvider subclass
- System Extension packaging (.systemext bundle)
- Embed helix-core as XCFramework in extension



Implement PacketTunnelProvider:

- startTunnel(options:completionHandler:) — init core, create TUN
- stopTunnel(with:completionHandler:) — cleanup, remove routes
- Handle IPC from main app via NETunnelProviderSession



Create Privileged Helper Tool (com.helixvpn.helper):

- XPC service for privileged operations (firewall, routes)
- Signed with same Team ID as main app
- SMJobBless installation on first launch



Configure entitlements:

- com.apple.developer.networking.networkextension
- com.apple.developer.system-extension.install
- App Sandbox with network-outgoing



Write macOS-specific PlatformAdapter implementation:

- TUN via NEPacketTunnelProvider.packetFlow
- Routes via
NEPacketTunnelProvider.setTunnelNetworkSettings()
- DNS via NETunnelNetworkSettings.DNSSettings
- apply_managed_policy() — parse macOS Configuration Profile (.mobileconfig) managed-preference keys pushed by helix-admin (forced kill switch, allowed protocols, DNS policy); cross-ref MVP2_ARCHITECTURE.md §10.2. End-to-end validation against a real MDM channel happens in Phase 9 §10.2 Week 32

Deliverables:

- macOS Network Extension building
- Privileged helper tool with XPC communication

- TUN working through NEPacketTunnelProvider
-

Week 8: macOS Kill Switch (PF) & Menu Bar

Tasks:



Implement macOS kill switch using Packet Filter (PF):

- Create PF anchor `com.helixvpn` for rule isolation
- Block all outbound traffic except VPN tunnel interface (`utun*`)
- Allow DHCP and VPN server endpoint
- Restore rules on disconnect
- Handle BFE (Base Filtering Engine) restart



Implement menu bar extra:

- `NSStatusItem` with VPN status icon
- Quick connect/disconnect from menu
- Server selection dropdown
- "Open Helix VPN" menu item



Implement keyboard shortcuts:

- `Cmd+Shift+C` — Connect/Disconnect toggle
- `Cmd+Shift+K` — Kill switch toggle



Auto-launch on login:

- `SMLaunchItemSetEnabled` for helper
- User preference persistence



macOS-specific UI polish:

- Native window chrome (title bar, traffic lights)
- Dark mode support (follows system preference)
- Notification Center integration for connection events

Deliverables:

- Kill switch functional on macOS (verified with leak test)
 - Menu bar extra with full functionality
 - Auto-launch and keyboard shortcuts
-

Week 9: Linux TUN + NetworkManager Integration

Tasks:



Implement Linux PlatformAdapter:

- TUN via `tun-rs` (already from Phase 1)
- Routing via `rtnetlink` crate (add/delete routes)

- DNS via D-Bus → systemd-resolved or NetworkManager
- Detect which DNS manager is active
- `apply_managed_policy()` — parse a self-hosted managed-config file drop (JSON/TOML watched via inotify) as the Linux equivalent of a Configuration Profile, since Linux has no single vendor-neutral MDM standard; cross-ref MVP2_ARCHITECTURE.md §10.2



Integrate with NetworkManager:

- Create VPN connection profile via D-Bus (`AddConnection`)
- Activate/deactivate via NetworkManager API
- Handle NM state change signals
- Graceful fallback to manual config if NM unavailable



Implement manual route management (fallback):

- `ip route` command execution for routing
- `/etc/resolv.conf` manipulation for DNS (with backup)
- Route table cleanup on disconnect



Linux system tray via `libappindicator`:

- Status icon showing VPN state
- Menu with connect/disconnect, server list, quit

Deliverables:

- Linux VPN working with NetworkManager
- Fallback manual mode functional
- System tray integration

Week 10: Linux Kill Switch (nftables) & Packaging

Tasks:



Implement Linux kill switch using nftables:

- Create nftables table `inet helix_vpn`
- Chain output with policy drop
- Rules: accept on `oifname "tun0"`, accept to VPN server, accept DHCP
- IPv6 leak prevention (drop all IPv6 outbound)
- Graceful cleanup on disconnect (flush table)



Fallback to iptables for legacy systems:

- `iptables -P OUTPUT DROP` + accept rules
- `ip6tables -P OUTPUT DROP` for IPv6



Package for distribution:

- **DEB** (Ubuntu 22.04+, Debian 12+): `dpkg-deb build`
- **RPM** (Fedora 39+, openSUSE): `rpmbuild spec`

- **Flatpak**: Flatpak manifest with freedesktop runtime
- **AppImage**: appimage-builder for universal Linux



Create .desktop file and app metadata



Implement OTA auto-update:

- Check for updates on launch (configurable)
- Download + signature verification
- Install with pkexec for privilege escalation

Deliverables:

- Kill switch on Linux (nftables + iptables fallback)
- DEB, RPM, Flatpak, AppImage packages
- OTA update mechanism

Milestone: M2 — Desktop Alpha (macOS + Linux)

Week 11: Core UI Implementation — Shared Desktop UI

Tasks:



Implement main UI layout (shared macOS/Linux):

- Connection screen (large connect button, status, current server)
- Server list screen (searchable, latency-sorted, favorites)
- Settings screen (protocol, kill switch, auto-connect, DNS)
- Statistics screen (real-time graph, session stats)



Implement connection state animations:

- Connecting spinner with progress steps
- Connected state with green indicator and location
- Disconnected state with gray indicator
- Error state with red indicator and retry button



Consume the existing OpenDesign token system — do NOT design a new palette/type-scale from scratch (OpenDesign is mandatory per MVP2_UI_UX_SPEC.md §1 and docs/design/README.md, already complete):

- Import docs/design/opendesign/helix/tokens.css custom properties into the Tauri/React frontend's Tailwind config
- Map OpenDesign color roles (primary, success, warning, error) and typography scale directly — no re-derivation
- Component library (buttons, inputs, toggles, cards) built against the imported tokens
- Dark/light mode theming driven by the token system's existing light+dark variants (system-follows per docs/design/README.md §2)



Accessibility compliance:

- ARIA labels on all interactive elements
- Keyboard navigation support
- Screen reader announcements for state changes

Deliverables:

- Complete desktop UI with all screens
 - OpenDesign tokens integrated into the desktop frontend (no bespoke design system re-derived)
 - Accessibility audit passed
-

Week 12: Desktop Polish & macOS Notarization

Tasks:



macOS app notarization setup:

- Apple Developer account configuration
- Code signing with Developer ID certificate
- xcrun notarytool integration in CI
- Staple notarization ticket to .dmg



Performance optimization:

- Bundle size audit (< 15 MB target)
- Memory usage optimization (< 80 MB idle target)
- Startup time optimization (< 1.5s target)



Desktop-specific features:

- Split tunneling UI (app list, rules)
- Protocol selection with latency display
- Connection log viewer (for support)
- Import/export WireGuard configuration



Wire the crash-reporting/telemetry SDK (Sentry, per Appendix D) behind an explicit privacy-consent opt-in prompt shown on first launch — no telemetry event fires before consent is recorded; this is the desktop half of the Phase 9 §10.2 cross-cutting "consent-gated telemetry pipeline" production-readiness milestone that must be live before any public beta



End-to-end desktop testing:

- Connect → verify IP change → disconnect → verify IP restored
- Kill switch test (disconnect network, verify block)
- DNS leak test (via ipleak.net API)
- Reconnection after sleep/wake

Deliverables:

- Notarized macOS .dmg ready for distribution
- Optimized Linux packages
- Full desktop test suite passing

Weeks 13-14: Desktop Integration Testing & Bug Fix

Tasks:



Cross-platform integration testing:

- Test matrix: macOS 12/13/14 × Linux Ubuntu/Fedora/Debian
- Automated UI tests with Tauri + Playwright
- Manual QA checklist execution



Bug triage and resolution:

- P0 bugs: block release (fix within 24h)
- P1 bugs: fix before beta
- P2 bugs: document for post-launch



Performance benchmarking:

- Connection establishment time (< 3s target)
- Throughput test (iperf3 via VPN tunnel)
- CPU usage during idle and active tunnel
- Memory leak detection (72-hour soak test)



Documentation:

- User guide (markdown + in-app help)
- Troubleshooting FAQ
- Changelog for alpha release

Deliverables:

- macOS + Linux desktop alpha builds
- Integration test report
- Performance benchmark results
- User documentation

4.3 Phase 3 Deliverables Summary

Deliverable	macOS	Linux	Notes
Tauri app	Yes	Yes	Shared codebase
NetworkExtension/TUN	Yes	Yes	Platform-specific
Kill switch	PF	nftables	

Deliverable	macOS	Linux	Notes
Menu bar / System tray	Yes	Yes	Both functional Platform-native
Packaging	.dmg	DEB/RPM/ Flatpak/ ApplImage	All signed
OTA updates	Yes	Yes	Shared mechanism
UI completeness	100%	100%	Shared React frontend

4.4 Phase 3 Exit Criteria

- ☐ macOS app notarized and installable
- ☐ Linux packages installable on Ubuntu 22.04, Fedora 39, Debian 12
- ☐ Kill switch functional on both platforms (verified with leak tests)
- ☐ Connect/disconnect cycle < 3 seconds
- ☐ No DNS leaks detected
- ☐ No P0 or P1 bugs outstanding
- ☐ All UI screens implemented and accessible
- ☐ `apply_managed_policy()` implemented for macOS Configuration Profile and Linux config-file-drop (real-channel validation deferred to Phase 9 §10.2)
- ☐ Crash-reporting/telemetry SDK wired and consent-gated (no events before opt-in)

Test Plan (cross-reference): Integration/connection tests per MVP2_SECURITY_PERFORMANCE.md §8.2; leak tests (DNS/WebRTC/IPv6) per §8.3; performance tests per §8.4; UI tests per §8.5; relevant success criteria: SC-06 (kill switch zero packet leakage), SC-12/SC-13/SC-14/SC-16/SC-17/SC-18 (bundle size, connection time, kill switch response, CPU/memory targets), SC-23/SC-24 (no DNS/IPv6 leaks) per MVP2_OVERVIEW.md §9.

5. Phase 4: Desktop Client — Windows (Weeks 12-18)

5.1 Phase Overview

Attribute	Value
Duration	7 weeks (overlaps with Phase 3 from Week 12)
Team	1 Desktop Developer (Windows specialist) + shared Rust Core
Depends On	Phase 3 UI patterns and core integration approach
Blocks	None (last desktop platform)
Goal	Full-featured Windows client with WFP kill switch and MSI installer

5.2 Week-by-Week Breakdown

Week 12: Windows Project Setup & WFP Research

Tasks:

- ☐ Set up Windows development environment:
 - Windows 11 + Visual Studio 2022 + Rust toolchain
 - WDK (Windows Driver Kit) for WFP development
 - windows-rs crate for Win32 API bindings
- ☐ Create Windows-specific Tauri build configuration:
 - tauri.conf.json with Windows-specific settings
 - WebView2 runtime detection and install prompt
 - Single-instance enforcement (tauri-plugin-single-instance)
- ☐ Research WFP integration patterns:
 - Windows Filtering Platform layer architecture
 - Callout driver registration
 - Filter weight positioning (avoid conflicts)
 - BFE (Base Filtering Engine) interaction
- ☐ Implement Windows PlatformAdapter skeleton:
 - setup_tunnel() — WinTUN adapter creation
 - configure_routes() — Windows routing table API
 - configure_dns() — SetInterfaceDnsSettings API
 - apply_managed_policy() — read Group Policy (ADMX template) / Intune-pushed configuration for forced kill switch, allowed protocols, DNS policy; cross-ref

MVP2_ARCHITECTURE.md §10.2. Real-channel validation happens in Phase 9 §10.2 Week 32

Deliverables:

- Windows dev environment documented
 - Tauri app building on Windows
 - WFP integration plan documented
-

Week 13: WinTUN Driver Integration

Tasks:



Integrate WinTUN driver:

- Bundle wintun.dll (signed by WireGuard Foundation)
- Load DLL at runtime and verify signature
- Create WinTUN adapter via WintunCreateAdapter
- Configure IP address, MTU, ring buffers



Implement Windows TUN read/write:

- Read packets from WinTUN ring buffer (async via overlapped I/O)
- Write packets to WinTUN ring buffer
- Handle adapter creation failures gracefully



Implement Windows routing:

- CreateIpForwardEntry for route management
- Add route to VPN server via default interface
- Redirect 0.0.0.0/0 to WinTUN adapter
- Route cleanup on disconnect (store original routes)



Implement Windows DNS configuration:

- SetInterfaceDnsSettings (Windows 10 19041+)
- Fallback to registry-based DNS (NameServer in registry)
- DNS restoration on disconnect

Deliverables:

- WinTUN adapter creation and packet I/O working
 - Routing configuration functional
 - DNS configuration functional
-

Week 14: Windows Service & System Tray

Tasks:



Implement Windows service wrapper:

- SERVICE_WIN32_OWN_PROCESS service type
- Service main entry point with SCM registration
- Service control handler (start, stop, pause, continue)
- Run core engine in service context



Implement service installer:

- CreateService with auto-start configuration
- Service account configuration (NT AUTHORITY\LocalService)
- Service removal on uninstall



Implement Windows system tray:

- Shell_NotifyIcon for tray icon
- Context menu: connect/disconnect, server list, settings, exit
- Balloon notifications for connection events
- Tray icon state changes (disconnected/connecting/connected)



Implement single-instance enforcement:

- Named mutex (Global\HelixVPN_SingleInstance)
- IPC to bring existing instance to foreground

Deliverables:

- Windows service installable and functional
- System tray with full menu
- Single-instance enforcement

Week 15: WFP Kill Switch Implementation

Tasks:



Implement WFP kill switch:

- Register WFP callout driver for outbound filtering
- Create persistent filters (survive BFE restart):
 - Block all outbound IPv4/IPv6 (weight: high priority)
 - Allow outbound on WinTUN adapter interface
 - Allow outbound to VPN server endpoint
 - Allow DHCP traffic (UDP 67/68)
 - Allow local network if split tunnel enabled
- Filter at FWPM_LAYER_ALE_AUTH_CONNECT_V4/V6



Handle WFP edge cases:

- BFE service restart detection and re-registration
- Filter conflict detection with other VPN products
- Clean filter removal on disconnect (no orphaned rules)
- Windows Firewall coexistence



Write kill switch tests:

- Functional test: connect → block → verify no traffic
- BFE restart test: simulate BFE stop/start → verify rules reinstalled
- Conflict test: install alongside other VPN → verify no crash

Deliverables:

- WFP kill switch fully functional
 - Persistent filters surviving BFE restart
 - Kill switch test suite passing
-

Week 16: MSI Installer & Windows Polish

Tasks:



Create MSI installer with WiX Toolset:

- Feature tree: Core app, WinTUN driver, Service, Start Menu shortcut
- Custom actions: install/remove Windows service
- WinTUN driver installation (.wintun device class)
- Per-machine installation (all users)
- Upgrade behavior (major upgrade)



Code signing:

- EV Code Signing Certificate
- Sign .exe, .dll, .msi artifacts
- SmartScreen reputation building



Windows-specific UI adjustments:

- Mica/Acrylic material backdrop (Windows 11)
- Title bar customization (centered title, custom buttons)
- Windows 10/11 visual consistency
- High contrast mode support



OTA updates for Windows:

- tauri-plugin-updater with Windows .msi/.nsis support
- Background download with UAC prompt for install
- Update channel support (stable, beta, alpha)



Wire crash-reporting/telemetry SDK (Sentry) behind the same privacy-consent opt-in prompt introduced in Phase 3 Week 12 — Windows half of the Phase 9 §10.2 consent-gated telemetry milestone

Deliverables:

- Signed MSI installer
- Windows service + WinTUN bundled

- OTA update mechanism

Milestone: M3 — Desktop Beta (All Desktop Platforms)

Weeks 17-18: Windows Testing & Bug Fix

Tasks:



Windows testing matrix:

- Windows 10 22H2 (x64)
- Windows 11 23H2 (x64)
- Windows 11 24H2 (x64, ARM64 if available)



Anti-virus compatibility testing:

- Windows Defender (default)
- Norton, McAfee, Kaspersky (common third-party)
- Verify WinTUN driver not flagged as malicious



Corporate environment testing:

- Active Directory domain join
- Group Policy restrictions
- Proxy server environments



Bug triage and fix sprint



Performance optimization for Windows:

- Startup time target: < 2s
- Memory target: < 80 MB idle
- Throughput target: > 500 Mbps on gigabit connection

Deliverables:

- Windows beta build ready for distribution
- AV compatibility report
- Performance benchmark results

5.3 Phase 4 Deliverables Summary

Deliverable	Status
Tauri app (Windows)	Shared with macOS/Linux
WinTUN driver integration	Signed driver bundled
Windows service	Auto-start, SCM registered
WFP kill switch	Persistent filters
System tray	Full menu + notifications
MSI installer	Signed, per-machine

Deliverable	Status
OTA updates	Plugin-based

5.4 Phase 4 Exit Criteria

- ☒ MSI installer installs and launches on Windows 10/11
- ☒ Kill switch functional (verified with leak test)
- ☒ Windows service starts automatically
- ☒ No AV false positives on WinTUN driver
- ☒ Connect/disconnect < 3 seconds
- ☒ No P0 or P1 bugs outstanding
- ☒ `apply_managed_policy()` implemented for Group Policy/Intune (real-channel validation deferred to Phase 9 §10.2)
- ☒ Crash-reporting/telemetry SDK wired and consent-gated

Test Plan (cross-reference): Integration/connection tests + leak tests per MVP2_SECURITY_PERFORMANCE.md §8.2/§8.3; performance tests per §8.4; security tests (AV/WFP conflict) per §8.6; relevant success criteria: SC-06, SC-13, SC-14, SC-15, SC-16, SC-17, SC-18, SC-23/SC-24 per MVP2_OVERVIEW.md §9.

6. Phase 5: Mobile — Android (Weeks 14-22)

6.1 Phase Overview

Attribute	Value
Duration	9 weeks (overlaps with Phase 4 from Week 14)
Team	2 Mobile Developers
Depends On	Phase 1 (core API), Phase 2 WireGuard (Week 4+)
Blocks	Phase 6 (iOS shares Flutter codebase)
Goal	Production-ready Android app with VpnService, split tunneling, and Play Store readiness

6.2 Week-by-Week Breakdown

Week 14: Flutter Project Setup & FRB Integration

Tasks:



Initialize Flutter 3.29+ project:

```
helix-mobile/
├── android/           # Android-specific code
├── ios/               # iOS-specific code
├── (placeholder)
├── lib/               # Dart shared code
│   ├── main.dart
│   ├── screens/       # UI screens
│   ├── bloc/          # State management (Bloc)
│   ├── services/      # Business logic
│   └── models/         # Data models
├── rust/              # Rust core
│   └── src/
└── pubspec.yaml
```



Integrate flutter_rust_bridge v2.0+:

- Add flutter_rust_bridge dependency to pubspec.yaml
- Configure frb_codegen build runner
- Create bridge API definitions in Rust
- Generate Dart bindings (flutter_rust_bridge_codegen generate)
- Verify bidirectional communication (Dart → Rust → Dart)



Set up state management with flutter_bloc:

- ConnectionBloc — VPN connection state
- ServerListBloc — Server discovery and selection
- SettingsBloc — User preferences
- StatsBloc — Real-time statistics



Configure Material 3 ThemeData as the consumption layer for the existing OpenDesign token system — do NOT derive a new palette/type-scale; OpenDesign is mandatory per MVP2_UI_UX_SPEC.md §1 (Android maps to "Material You + Dynamic" per docs/design/README.md §2):

- Generate ColorScheme/TextTheme from the OpenDesign token export (light + dark)
- Typography mapped from the OpenDesign type scale onto Material 3's type roles
- Custom components (connection button, server card, stat gauge) styled from the imported tokens, not ad-hoc values

Deliverables:

- Flutter app building for Android
 - FRB bridge functional (ping-pong test)
 - Bloc state management scaffold
 - OpenDesign tokens integrated into Material 3 ThemeData (no bespoke design system re-derived)
-

Week 15: Android VpnService Implementation

Tasks:



Implement HelixVpnService extending `android.net.VpnService`:

- `onStartCommand()` — handle connect/disconnect intents
- `onCreate()` — initialize Rust core via FRB
- `onDestroy()` — cleanup



Implement Builder configuration:

- `addAddress()` — tunnel IP assignment
- `addRoute()` — 0.0.0.0/0 for full tunnel
- `addDnsServer()` — VPN DNS servers
- `setMtu()` — 1420 for WireGuard
- `setBlocking(true)` — kill switch on Android
- `establish()` — create TUN file descriptor



Implement socket protection:

- `protect(socketFd)` for WireGuard UDP transport socket
- Prevents VPN tunnel traffic from looping back into tunnel



Implement Android PlatformAdapter:

- TUN via `VpnService.Builder.establish()` (returns `ParcelFileDescriptor`)
- Routes via `Builder.addRoute()`
- DNS via `Builder.addDnsServer()`
- Kill switch via `Builder.setBlocking(true)`
- `apply_managed_policy()` — consume Android Enterprise managed configuration (`RestrictionsManager / AppConfig` schema) pushed by an EMM console via `helix-admin`; cross-ref MVP2_ARCHITECTURE.md §10.2. Real-channel validation happens in Phase 9 §10.2 Week 32



Pass TUN file descriptor to Rust core:

- Convert `ParcelFileDescriptor` to raw fd
- Send fd via FRB to Rust `UnixTun` implementation

Deliverables:

- VpnService functional (TUN created, fd passed to core)
- Basic connect/disconnect working on Android

- Socket protection preventing loops
-

Week 16: Foreground Service & Quick Settings Tile

Tasks:



Implement foreground service:

- `startForeground(notificationId, notification)` with `Notification.Builder`
- `FOREGROUND_SERVICE_TYPE_SPECIAL_USE` (Android 14+)
- Persistent notification showing VPN status
- Notification actions: Disconnect, Change Server
- Handle foreground service restrictions (Android 12+ bg start limits)



Implement Quick Settings tile:

- `HelixVpnTileService` extending `TileService`
- Tile state: `STATE_INACTIVE` (disconnected), `STATE_ACTIVE` (connected)
- Tap to toggle connection
- Long-press to open app
- Icon changes based on state



Implement notification channel:

- `NotificationChannel` with low importance (no sound/vibration)
- Status icon showing connected/disconnected state
- Expandable notification with stats (duration, data used)



Handle Android lifecycle edge cases:

- Doze mode / App Standby (keepalive handling)
- Battery saver mode (reduce keepalive frequency)
- VPN killed by system (auto-reconnect if enabled)
- Device reboot (auto-start if enabled)

Deliverables:

- Foreground service with persistent notification
 - Quick Settings tile functional
 - Lifecycle edge cases handled
-

Week 17: Split Tunneling & App Management

Tasks:



Implement app list retrieval:

- Query PackageManager for installed apps
- Filter user-installed vs. system apps
- Retrieve app icons and labels
- Cache app list with refresh capability



Implement split tunneling UI:

- App list with toggle switches (include/exclude)
- Search and filter functionality
- Categories (browsers, streaming, gaming, etc.)
- Preset profiles ("Streaming", "Gaming", "Work")



Implement per-app VPN rules:

- Builder.addAllowedApplication() — include mode
- Builder.addDisallowedApplication() — exclude mode
- Mode toggle: include only selected / exclude selected
- Rule persistence across reconnects



Implement per-app connection stats:

- Track bytes per app (via TrafficStats or kernel counters)
- Display in UI with progress bars
- Export usage report

Deliverables:

- Split tunneling with app selection
 - Per-app stats tracking
 - Preset profiles
-

Week 18: Battery Optimization & Deep Integration

Tasks:



Implement battery optimization:

- Request REQUEST_IGNORE_BATTERY_OPTIMIZATIONS permission
- Guide user to whitelist app in system settings
- Adaptive keepalive (shorter on WiFi, longer on mobile data)
- Reduce packet batching on battery saver



Implement connection resilience:

- Network change detection
(ConnectivityManager.NetworkCallback)
- Auto-reconnect on network switch (WiFi → mobile → WiFi)
- Roaming detection and user confirmation
- Captive portal detection and bypass



Implement Android shortcuts:

- App shortcuts (long-press launcher icon)

- Connect to favorite server
- Toggle kill switch



Implement widget (Android 12+):

- Home screen widget showing connection status
- One-tap connect/disconnect
- Current server display



Wire crash-reporting/telemetry SDK (Sentry) behind an explicit privacy-consent opt-in prompt (Play Store Data Safety form declares this honestly) — mobile half of the Phase 9 §10.2 consent-gated telemetry milestone that must be live before any public beta

Deliverables:

- Battery optimization with user guidance
- Auto-reconnect on network changes
- Shortcuts and widget

Week 19: Core Mobile UI Implementation

Tasks:



Implement connection screen:

- Large animated connection button (ripple/pulse animation)
- Current server display with flag
- Connection duration timer
- Real-time speed graph (download/upload)
- Quick server change dropdown



Implement server list screen:

- Country/region grouping with expandable sections
- Latency indicators (color-coded: green < 50ms, yellow < 150ms, red > 150ms)
- Favorite/bookmark functionality
- Search with autocomplete
- Recently used servers
- Recommended server (lowest latency)



Implement settings screen:

- Protocol selection (WireGuard, Shadowsocks)
- Kill switch toggle
- Auto-connect preferences (on boot, on untrusted WiFi)
- DNS settings (auto, custom)
- Split tunneling link
- About section (version, licenses)



Implement account screen:

- Subscription status display
- Data usage (monthly/quota)
- Logout functionality

Deliverables:

- All primary UI screens implemented
 - Animations and transitions
 - Responsive layout (phone + tablet)
-

Week 20: Android Testing & Play Store Preparation

Tasks:



Android testing matrix:

- Devices: Pixel 7 (Android 14), Samsung Galaxy S23 (Android 14), Xiaomi 13 (Android 13)
- Emulators: API 26, 30, 33, 34 (various screen sizes)
- Architecture: ARM64, x86_64



Automated testing:

- Widget tests with `flutter_test`
- Integration tests with `integration_test` driver
- FRB bridge unit tests



Play Store preparation:

- Create Google Play Console listing
- Screenshots (phone + tablet, light + dark mode)
- Feature graphic and app icon
- Privacy policy compliance
- App content rating (ESRB/PEGI)
- Data safety form
- Target API level compliance (API 34+)



Build AAB (Android App Bundle) for Play Store:

- ABI split (ARM64, ARMv7, x86_64)
- Screen density split
- Language split

Deliverables:

- Tested Android APK/AAB
- Play Store listing ready for submission
- Automated test suite

Milestone: M4 — Mobile Alpha (Android)

Weeks 21-22: Android Polish & Bug Fix Sprint

Tasks:



OEM-specific compatibility testing:

- Samsung (One UI battery optimization)
- Xiaomi (MIUI background restrictions)
- Huawei (App Launch restrictions)
- Oppo/Vivo (ColorOS/FuntouchOS)



Bug triage and fix sprint:

- P0: crash on specific devices, connection failures
- P1: UI glitches, performance issues
- P2: feature requests, minor polish



Performance optimization:

- App startup time < 2s
- APK size < 25 MB
- Memory usage < 120 MB at idle
- Battery drain < 3% per hour (active VPN)



Final QA sign-off

Deliverables:

- Android beta build ready
- OEM compatibility report
- Performance benchmarks

6.3 Phase 5 Deliverables Summary

Deliverable	Status
Flutter app with Material 3	Complete
VpnService integration	Functional
Foreground service	With notification
Quick Settings tile	Toggle functional
Split tunneling	Per-app rules
Battery optimization	Whitelist + adaptive
Auto-reconnect	Network change aware
Widget + shortcuts	Implemented
Play Store ready	Listing prepared

6.4 Phase 5 Exit Criteria



- ☐ VPN connects and routes traffic on Android 12-14
- ☐ Kill switch functional via `setBlocking(true)`
- ☐ Split tunneling with app selection works
- ☐ Foreground service persistent across device sleep
- ☐ Quick Settings tile functional
- ☐ APK/AAB size < 25 MB
- ☐ No P0 or P1 bugs on target devices
- ☐ `apply_managed_policy()` implemented for Android Enterprise managed configuration (real-channel validation deferred to Phase 9 §10.2)
- ☐ Crash-reporting/telemetry SDK wired and consent-gated (declared honestly in the Play Store Data Safety form)

Test Plan (cross-reference): Integration tests per MVP2_SECURITY_PERFORMANCE.md §8.2; performance/battery tests per §8.4/§5.5; UI tests per §8.5; relevant success criteria: SC-01, SC-07 (split tunneling), SC-08 (auto-connect < 5s), SC-12/SC-18/SC-19 (APK size, memory, battery impact) per MVP2_OVERVIEW.md §9.

7. Phase 6: Mobile — iOS (Weeks 20-26)

7.1 Phase Overview

Attribute	Value
Duration	7 weeks (overlaps with Phase 5 from Week 20)
Team	1 Mobile Developer (iOS specialist) + shared Flutter dev
Depends On	Phase 5 (shared Flutter codebase), Phase 1 (UniFFI bindings)
Blocks	None (last mobile platform)
Goal	iOS app with NEPacketTunnelProvider, memory optimization, and App Store readiness

7.2 Week-by-Week Breakdown

Week 20: iOS Flutter Setup & UniFFI Swift Bindings

Tasks:



Configure iOS Flutter project:

- Set minimum iOS version to 15.0
- Configure Xcode project settings
- Set up CocoaPods / Swift Package Manager
- Configure signing and provisioning (development)



Integrate UniFFI-generated Swift bindings:

- Generate Swift bindings from `helix_core.udl`
- Create `HelixCoreFFI` Swift package
- Bridge UniFFI types to Dart via FRB + platform channel
- Verify Rust function calls from Swift



Implement iOS platform channel bridge:

- Dart `MethodChannel` for iOS-specific operations
- Swift handler for platform channel calls
- Bidirectional event streaming (state changes to Dart)



Set up iOS app group for IPC:

- `group.com.helixvpn.app` shared container
- Keychain sharing between app and extension
- Shared preferences via `UserDefaults(suiteName:)`

Deliverables:

- iOS Flutter app building and running on simulator
 - UniFFI Swift bindings integrated
 - Platform channel bridge functional
-

Week 21: NEPacketTunnelProvider Extension

Tasks:



Create Network Extension target:

- `HelixPacketTunnel` — `NEPacketTunnelProvider` subclass
- Bundle identifier: `com.helixvpn.app.tunnel`
- Target iOS 15.0+



Implement `PacketTunnelProvider`:

```
class PacketTunnelProvider: NEPacketTunnelProvider {  
    private var core: HelixVpnCore?
```

```

override func startTunnel(
    options: [String: NSObject]?,
    completionHandler: @escaping (Error?) -> Void
) {
    // Load config from shared defaults
    // Initialize Rust core via UniFFI
    // Start WireGuard handshake
    // Configure tunnel network settings
}

override func stopTunnel(
    with reason: NEProviderStopReason,
    completionHandler: @escaping () -> Void
) {
    // Graceful disconnect
    // Cleanup resources
}
}

```



Implement tunnel network settings:

- NEPacketTunnelNetworkSettings with:
 - NEIPv4Settings — tunnel address, routes
 - NEDNSSettings — VPN DNS servers
 - NEProxySettings — proxy if needed
- setTunnelNetworkSettings() for atomic application



Handle packet flow:

- packetFlow.readPackets() — read IP packets from TUN
- packetFlow.writePackets() — write decrypted packets to TUN
- Bridge to Rust core via TunnelDevice trait implementation



Request Network Extension entitlement from Apple:

- Submit entitlement request via Apple Developer portal
- Include privacy policy and app description
- Wait for approval (may take 1-2 weeks — start early)



apply_managed_policy() — consume Apple MDM managed app configuration (per-app VPN NEVPNProtocol managed settings + com.apple.app.managed configuration dictionary) pushed by an MDM console via helix-admin; cross-ref MVP2_ARCHITECTURE.md §10.2. Real-channel validation happens in Phase 9 §10.2 Week 32

Deliverables:

- Network Extension building and loading
- Packet flow bridged to Rust core
- Tunnel network settings applied

- `apply_managed_policy()` consuming Apple MDM managed app config
-

Week 22: Memory Optimization (< 15MB)

Tasks:



Profile memory usage:

- Xcode Memory Graph Debugger
- Instruments → Allocations
- Target: < 15 MB for Network Extension (iOS limit)



Optimize Rust core for iOS constraints:

- Reduce Tokio worker threads (2 instead of 4)
- Limit channel buffer sizes (128 instead of 1024)
- Use `opt-level = "z"` and `lto = true`
- Strip all symbols
- Target binary size: < 5 MB for core .a



Optimize packet handling:

- Use `bytes::Bytes` for zero-copy (already done)
- Reuse packet buffers instead of allocating per-packet
- Batch packet reads/writes where possible



Manage extension lifecycle:

- Aggressive cleanup on disconnect (drop all resources)
- Minimize memory during idle (no pre-allocated buffers)
- Use `autoreleasepool` in Swift for tight loops



Optimize Flutter memory:

- Reduce image asset sizes
- Lazy-load server list (paginate)
- Cache management (LRU for flags/icons)

Deliverables:

- Memory usage < 15 MB in Network Extension
 - Core binary < 5 MB
 - Memory optimization report
-

Week 23: On-Demand Rules & iOS Deep Integration

Tasks:



Implement VPN On-Demand rules:

- NEOnDemandRuleConnect — auto-connect on specific WiFi SSIDs
- NEOnDemandRuleDisconnect — auto-disconnect on trusted networks
- NEOnDemandRuleEvaluateConnection — conditional connect
- User-configurable trusted/untrusted network lists



Implement iOS-specific features:

- Siri Shortcuts integration (Intents framework)
 - "Connect Helix VPN" voice command
 - "Disconnect Helix VPN" voice command
- Widget (iOS 14+):
 - Home screen widget with connection status
 - Tap to toggle
- Live Activity (iOS 16+):
 - Dynamic Island / Lock Screen VPN status
 - Real-time speed display



Handle iOS-specific lifecycle:

- Background fetch for server list updates
- Background processing for keepalive
- Low Power Mode adaptation (reduce keepalive)
- iPad multitasking support (Split View, Slide Over)

Deliverables:

- On-demand rules functional
- Siri Shortcuts integration
- Widget and Live Activity

Week 24: iOS Testing & App Store Preparation

Tasks:



iOS testing matrix:

- iPhone 14 Pro (iOS 17)
- iPhone 13 (iOS 16)
- iPhone SE 3rd gen (iOS 15) — smallest screen
- iPad Pro 11" (iPadOS 17)
- Simulators: iOS 15, 16, 17



App Store preparation:

- Create App Store Connect record
- Screenshots (iPhone 6.7", 6.5", 5.5", iPad Pro 12.9", iPad Pro 11")
- App Preview video (15-30 seconds)
- App icon (all required sizes)
- Privacy policy URL

- App Review information (demo account, notes)
- Export compliance documentation (uses standard encryption)



Beta distribution:

- TestFlight internal testing (team)
- TestFlight external testing (beta users)
- Build upload via Xcode / altool



Network Extension entitlement follow-up:

- Ensure entitlement approved by Apple
- Verify provisioning profile includes entitlement

Deliverables:

- TestFlight build uploaded
- App Store listing prepared
- All device testing passed

Milestone: M5 — Mobile Beta (Android + iOS)

Weeks 25-26: iOS Polish & Cross-Mobile Bug Fix

Tasks:



iOS-specific polish:

- Cupertino-style UI adaptation (iOS-specific widgets)
- Dynamic Type support (accessibility text sizes)
- VoiceOver support (screen reader labels)
- Haptic feedback on connection state changes



Wire crash-reporting/telemetry SDK (Sentry) behind an explicit privacy-consent opt-in prompt, App Tracking Transparency-compliant where applicable — iOS half of the Phase 9 §10.2 consent-gated telemetry milestone that must be live before any public beta (TestFlight)



Cross-platform mobile bug fix:

- Shared Flutter code issues
- Platform-specific regressions
- Performance issues on lower-end devices



Final QA:

- Full regression test on Android and iOS
- Battery drain comparison (with/without VPN)
- Speed test comparison (native speed vs. VPN speed)
- Long-running stability test (24-hour tunnel)

Deliverables:

- iOS beta build
- Cross-platform mobile QA report
- Performance comparison data

7.3 Phase 6 Deliverables Summary

Deliverable	Status
Flutter app (iOS)	Shared with Android
NEPacketTunnelProvider	Functional
Memory optimization	< 15 MB extension
On-demand rules	WiFi-based auto-connect
Siri Shortcuts	Voice commands
Widget / Live Activity	iOS 14+ / 16+
TestFlight ready	Build uploaded

7.4 Phase 6 Exit Criteria



VPN connects on iOS 15-17



Network Extension memory < 15 MB



On-demand rules functional



TestFlight build uploaded



No P0 or P1 bugs



App Store listing complete



apply_managed_policy() implemented for Apple MDM managed app config (real-channel validation deferred to Phase 9 §10.2)



Crash-reporting/telemetry SDK wired and consent-gated before TestFlight distribution

Test Plan (cross-reference): Integration tests per MVP2_SECURITY_PERFORMANCE.md §8.2; performance/memory tests per §8.4; UI tests (VoiceOver/Dynamic Type) per §8.5; relevant success criteria: SC-01, SC-09 (biometric auth), SC-12/SC-18 (bundle size, memory <15 MB extension) per MVP2_OVERVIEW.md §9.

8. Phase 7: HarmonyOS & Aurora OS (Weeks 24-30)

8.1 Phase Overview

Attribute	Value
Duration	7 weeks (overlaps with Phase 6 from Week 24)
Team	1 HarmonyOS Specialist
Depends On	Phase 5-6 (mobile patterns), Phase 1 (core)
Blocks	None
Goal	HarmonyOS and Aurora OS clients with distribution readiness

8.2 Week-by-Week Breakdown

Week 24: HarmonyOS Flutter (ohos) Build Setup

Tasks:



Set up HarmonyOS development environment:

- DevEco Studio (based on IntelliJ IDEA)
- OpenHarmony SDK (API 12+)
- Flutter v3.22.0-ohos embedding
- Rust aarch64-unknown-linux-ohos target



Configure Flutter for HarmonyOS build:

- Add ohos platform to Flutter project
- Configure flutter build hap build path
- Set up build-profile.json5 for HarmonyOS
- Configure signing with debug certificate



Build helix-core for HarmonyOS target:

- Configure cross with OHOS SDK linker
- Compile with --target aarch64-unknown-linux-ohos
- Verify binary runs on HarmonyOS emulator



Test FRB on HarmonyOS:

- Verify bridge communication
- Handle any platform-specific FFI issues

Deliverables:

- HarmonyOS dev environment documented
- helix-core compiling for OHOS target

- Basic Flutter app running on HarmonyOS emulator
-

Week 25: VpnExtensionAbility Integration

Tasks:



Implement HarmonyOS VPN extension:

- Create VpnExtensionAbility subclass
- Implement onCreate(), onConnect(), onDisconnect()
- Handle VPN configuration via VpnConfig



Implement HarmonyOS PlatformAdapter:

- TUN via HarmonyOS VPN APIs (VpnExtensionAbility.setTunnel())
- Routing via HarmonyOS network management
- DNS via system DNS configuration APIs
- Kill switch via setBlocking(true) equivalent



Implement N-API bridge for direct Rust calls:

- Create Native API (N-API) module in HarmonyOS
- Bind to Rust core functions via FFI
- Handle type marshalling between ArkTS/TS and Rust



Handle HarmonyOS-specific constraints:

- Background execution limitations
- Power management (no foreground service equivalent)
- Network permission model

Deliverables:

- VpnExtensionAbility functional
 - N-API bridge to Rust core
 - Basic connect/disconnect on HarmonyOS
-

Week 26: HarmonyOS UI & Feature Parity

Tasks:



Adapt Flutter UI for HarmonyOS:

- HarmonyOS design language adaptation
- ArkUI-style components where appropriate
- Screen size adaptation (phones, tablets, foldables)



Implement HarmonyOS-specific features:

- Service widget (HarmonyOS home screen widget)
- Atomic service (HarmonyOS quick app)

- Distributed capability (cross-device VPN)
- ☐

Feature parity with Android:

- Connect/disconnect, server list, settings
- Kill switch, split tunneling (if supported)
- Auto-connect preferences



Testing on HarmonyOS emulator:

- Functional testing
- Performance profiling
- Memory usage verification

Deliverables:

- HarmonyOS UI complete
 - Feature parity with Android
 - .hap package building
-

Week 27: Aurora OS Qt Project Setup

Tasks:



Set up Aurora OS development environment:

- Aurora SDK (Sailfish SDK fork)
- Qt 6.7+ / QML toolchain
- GCC cross-compiler for ARM64
- Rust aarch64-unknown-linux-gnu target



Initialize Qt6/QML project:

```
helix-aurora/
├── src/
│   ├── main.cpp           # Qt application entry
│   ├── main.qml           # Root QML
│   ├── vpncontroller.cpp  # C++ bridge to Rust
│   └── vpncontroller.h
├── qml/
│   ├── pages/             # QML pages
│   ├── components/       # Reusable QML components
│   └── styles/            # Silica styling
├── rust/
│   └── src/               # Rust core FFI bindings
└── rpm/                  # RPM packaging
```

└── helix-aurora.pro

- [] Configure `cbindgen` for C header generation:
- Generate `helix\_core.h` from Rust public API
- Include in C++ build

- Verify function signatures match
- [] Implement C++ bridge (`VpnController`):
- QObject-derived class with Q_PROPERTY for state
- Slots: `connect()`, `disconnect()`, `selectServer()`
- Signals: `connectionStateChanged()`, `statsUpdated()`
- Thread-safe Rust calls via QtConcurrent

****Deliverables:****

- Aurora OS dev environment documented
- Qt project building
- C++ bridge to Rust functional

Week 28: ConnMan VPN Plugin & Silica UI

****Tasks:****

- [] Implement ConnMan VPN plugin integration:
- Create D-Bus service at `net.connman.vpn`
- Implement `net.connman.vpn.Agent` interface
- Handle VPN configuration via ConnMan API
- Integrate with Aurora OS network settings
- [] Implement Silica UI components:
- `Page` for main connection screen
- `PullDownMenu` for server selection
- `Label` + `BusyIndicator` for connection state
- `Slider` for settings (MTU, etc.)
- `Switch` for kill toggle, auto-connect
- [] Implement Silica-specific UX:
- Peek gesture for page navigation
- Cover page (active frame) showing VPN status
- Ambience integration (theme following system)
- Sailfish-style pulley menus
- [] Handle Aurora OS lifecycle:
- Background execution (no foreground service)
- Device sleep handling
- Network state changes via ConnMan signals

****Deliverables:****

- ConnMan VPN plugin functional
- Silica UI complete
- Cover page with status

****Milestone: M6 – Extended Platforms (HarmonyOS + Aurora OS)****

Weeks 29-30: Distribution Preparation & Testing

****Tasks:****

- [] HarmonyOS AppGallery preparation:
- Create Huawei AppGallery developer account

- Prepare app listing (screenshots, description)
- Build signed release ``.hap``
- Submit for review
- [] Aurora OS OpenRepos preparation:
 - Create OpenRepos developer account
 - Build signed RPM package
 - Write Chum (Sailfish package manager) metadata
 - Submit to OpenRepos
- [] Cross-platform testing:
 - HarmonyOS: emulator + physical device (if available)
 - Aurora OS: SDK emulator
 - Functional regression testing
 - Performance comparison with reference platforms
- [] Documentation:
 - Platform-specific build instructions
 - Installation guides
 - Known limitations document

****Deliverables:****

- HarmonyOS ``.hap`` submitted to AppGallery
- Aurora OS RPM on OpenRepos
- Platform-specific documentation

8.3 Phase 7 Deliverables Summary

Deliverable	HarmonyOS	Aurora OS
Platform adapter	VpnExtensionAbility	ConnMan VPN plugin
UI framework	Flutter (ohos)	Qt6/QML (Silica)
Rust bridge	N-API + FRB	Direct C FFI
Package format	``.hap``	``.rpm``
Distribution	AppGallery	OpenRepos
Kill switch	Limited	nftables

8.4 Phase 7 Exit Criteria

- [x] HarmonyOS app connects and routes traffic (WireGuard, verified via ``ipleak.net``-equivalent IP-change check on emulator/device)
- [x] Aurora OS app integrates with ConnMan (D-Bus ``net.connman.vpn.Agent`` registered, connect/disconnect verified via ConnMan CLI + traffic capture)
- [x] Both apps submitted to respective stores (HarmonyOS AppGallery, Aurora OS OpenRepos) with no submission-blocking review feedback outstanding
- [x] Feature parity checklist explicitly enumerated and verified per platform (not merely asserted): connect/disconnect, server list, settings, auto-connect – present on both; kill switch – HarmonyOS "Limited"

(documented
constraint per §8.3 table), Aurora OS via nftables (full); split
tunneling –
documented as unsupported on HarmonyOS/Aurora OS if the platform
API lacks
the primitive (honest gap per SC-07's Tier-1 scope, not silently
implied)
- [x] `apply\_managed\_policy()` – HarmonyOS implements it "where
available" via
HarmonyOS enterprise device management (best-effort, cross-ref
`MVP2\_ARCHITECTURE.md` §10.2); Aurora OS is explicitly OUT OF
SCOPE for MVP2
managed-policy push (niche/single-device deployments per
`MVP2\_ARCHITECTURE.md` §10.5) – documented, not silently omitted
- [x] No P0 bugs

****Test Plan (cross-reference):**** Integration/connection tests per
`MVP2\_SECURITY\_PERFORMANCE.md` §8.2 (dispatched to the HarmonyOS/
Aurora OS
topology per §11.4.3-class per-environment test dispatch); manual
QA per
`MVP2\_SECURITY\_PERFORMANCE.md` §8.5 where the emulator lacks
physical-device
parity; relevant success criteria: SC-01 (platform functional),
SC-06 (kill
switch, Aurora OS only) per `MVP2\_OVERVIEW.md` §9.

9. Phase 8: Web & Browser Extension (Weeks 28-34)

9.1 Phase Overview

Attribute	Value
Duration	7 weeks (overlaps with Phase 7 from Week 28)
Team	1 Web Developer
Depends On	Phase 1 (WASM crypto build), Phase 2 (protocol config)
Blocks	None
Goal	Browser extension (MV3), native messaging host, web admin panel, and PWA companion

9.2 Week-by-Week Breakdown

Week 28: Browser Extension Scaffold (MV3)

****Tasks:****

- [] Initialize Manifest V3 browser extension:

```

helix-web/ └─ manifest.json # MV3 manifest └─ src/ └─
background.ts # Service worker └─ └─ popup/ # Popup UI └─ └─
popup.html └─ └─ popup.tsx └─ └─ popup.css └─ └─ content/ #
Content scripts └─ └─ options/ # Options page └─ └─ shared/ #
Shared utilities └─ └─ wasm/ # WASM crypto module └─ └─ public/ #
Static assets └─ └─ package.json

```

- [] Configure MV3 manifest:

```

```json
{
 "manifest_version": 3,
 "name": "Helix VPN",
 "version": "2.0.0",
 "permissions": [
 "proxy",
 "storage",
 "nativeMessaging"
],
 "host_permissions": [
 "<all_urls>"
],
 "background": {
 "service_worker": "background.js"
 },
 "action": {
 "default_popup": "popup.html"
 }
}

```



Implement service worker:

- Extension lifecycle management
- Proxy configuration handler
- Native messaging host management
- State persistence via `chrome.storage`



Implement popup UI:

- Connection toggle button
- Status display (connected/disconnected)
- Server selection (simplified list)
- Link to full settings

## Deliverables:

- MV3 extension building
  - Service worker functional
  - Popup UI rendering
-

## Week 29: Proxy Mode & PAC Script

### Tasks:



Implement proxy configuration:

- Generate PAC (Proxy Auto-Configuration) script dynamically
- `chrome.proxy.settings.set()` with `pacScript`
- Route browser traffic through VPN gateway proxy
- Bypass proxy for local addresses



Implement PAC script logic:

```
function FindProxyForURL(url, host) {
 // Direct for local addresses
 if (isPlainHostName(host) || shExpMatch(host, "*.local")) {
 return "DIRECT";
 }
 // VPN proxy for everything else
 if (helixVpnConnected) {
 return "PROXY <vpn-gateway>:<proxy-port>";
 }
 return "DIRECT";
}
```



Implement proxy authentication:

- `chrome.webRequest.onAuthRequired` handler
- Credential retrieval from secure storage
- Session token management



Handle proxy edge cases:

- WebSocket upgrade through proxy
- DNS resolution via proxy (prevent DNS leaks)
- QUIC/HTTP3 fallback handling

### Deliverables:

- Proxy mode routing browser traffic
  - PAC script generation
  - Authentication working
- 

## Week 30: Native Messaging Host

### Tasks:



Implement native messaging host:

- Standalone executable (`helix-nm-host`)

- STDIN/STDOUT JSON message protocol
- Message types: connect, disconnect, status, get\_servers
- ☐ Register native messaging host:
  - com.helixvpn.nm.json manifest
  - Browser registry / plist registration
  - Path to host executable
- ☐ Bridge to desktop app:
  - Host executable delegates to helix-desktop via IPC
  - Fallback to standalone mode (if desktop app not installed)
  - Graceful degradation messaging
- ☐ Implement native messaging protocol:

```
// Extension → Host
{"type": "connect", "server_id": "us-east-1", "protocol":
 "wireguard"}

// Host → Extension
{"type": "status", "connected": true, "server": "us-east-1",
 "ip": "10.64.0.1"}
```

### Deliverables:

- Native messaging host functional
  - Extension ↔ Desktop app communication
  - Graceful fallback when desktop not installed
- 

## Week 31: WASM Crypto Module

### Tasks:

- ☐ Compile Rust crypto to WASM:
  - wasm-pack build --target web for helix-crypto
  - wasm-bindgen for JS glue generation
  - Target: wasm32-unknown-unknown
- ☐ Implement WASM crypto API:
  - generate\_keypair() — X25519 keypair generation
  - derive\_shared\_secret(private\_key, public\_key) — X25519 DH
  - encrypt\_chacha20poly1305(key, nonce, plaintext) — AEAD encrypt
  - decrypt\_chacha20poly1305(key, nonce, ciphertext) — AEAD decrypt
- ☐ Integrate WASM into extension:
  - Load WASM module in service worker
  - Use for key generation and local crypto operations
  - Performance: key generation < 10ms in WASM





Handle WASM limitations:

- No raw socket access (enforced by browser)
- 4GB memory limit
- Single-threaded (no SharedArrayBuffer needed for crypto)

### **Deliverables:**

- WASM crypto module building
  - Key generation and encryption in browser
  - Performance benchmarks
- 

## **Week 32: Web Admin Panel & PWA**

### **Tasks:**



Implement web admin panel (React):

- Device management (list connected devices)
- Server management (add/remove/edit servers)
- User management (accounts, subscriptions)
- Analytics dashboard (connections, bandwidth, errors)
- Configuration deployment (push settings to clients)



Implement PWA companion:

- manifest.json for PWA installability
- Service worker for offline capability
- Connection status display (syncs with extension)
- Push notifications for security alerts



Implement admin API client:

- REST API integration (/api/v2/admin/\*)
- Authentication (JWT tokens)
- Role-based access control



Admin security:

- MFA (TOTP) support
- Session management
- Audit logging



apply\_managed\_policy() equivalent for the browser extension — read Chrome chrome.storage.managed (populated by ExtensionSettings/ExtensionInstallForcelist policy) and Firefox policies.js on for forced protocol/DNS/kill-switch policy pushed by an enterprise admin; cross-ref MVP2 ARCHITECTURE.md §10.2. Real-channel validation happens in Phase 9 §10.2 Week 32



Wire crash-reporting/telemetry SDK (Sentry) into the service worker + web admin panel behind an explicit privacy-consent

opt-in prompt — web half of the Phase 9 §10.2 consent-gated telemetry milestone

### **Deliverables:**

- Web admin panel functional
- PWA installable and functional
- Admin API integration
- Browser-extension managed-policy consumption (Chrome/Firefox enterprise policy channels)
- Crash-reporting/telemetry wired and consent-gated

### **Milestone: M7 — Web Complete (Extension + Admin)**

---

## **Weeks 33-34: Cross-Browser Testing & Store Submissions**

### **Tasks:**



Cross-browser testing:

- Chrome 120+ (primary)
- Firefox 121+ (MV3 support)
- Edge 120+ (Chromium-based)
- Safari 17+ (limited MV3 support — may need adjustments)



Browser-specific adaptations:

- Firefox: `browser.*` API namespace (polyfill)
- Safari: `browser.webRequest` limitations
- Chrome: full MV3 feature set



Store submissions:

- Chrome Web Store: zip upload, listing, screenshots
- Firefox Add-ons (AMO): xpi upload, listing
- Edge Add-ons: submission via Partner Center
- Safari: Mac App Store (if supporting Safari)



Extension testing:

- Proxy mode functional in all browsers
- Native messaging in Chrome/Edge
- Performance: popup opens in < 100ms
- Memory: service worker < 50 MB

### **Deliverables:**

- Extension submitted to Chrome Web Store, AMO, Edge Add-ons
- Cross-browser compatibility report
- PWA deployed

## 9.3 Phase 8 Deliverables Summary

Deliverable	Chrome	Firefox	Edge	Safari
Manifest V3 extension	Yes	Yes	Yes	Limited
Proxy mode (PAC)	Yes	Yes	Yes	Yes
Native messaging	Yes	Yes	Yes	No
WASM crypto	Yes	Yes	Yes	Yes
Web admin panel	Yes	Yes	Yes	Yes
PWA	Yes	Yes	Yes	Yes

## 9.4 Phase 8 Exit Criteria

- ☒ Extension routes browser traffic through proxy
- ☒ Native messaging works with desktop app
- ☒ WASM crypto functional and performant
- ☒ Admin panel accessible and functional
- ☒ PWA installable
- ☒ Submitted to at least 2 browser stores
- ☒ Managed-policy consumption implemented for Chrome ExtensionSettings/ ExtensionInstallForcelist and Firefox policies.json (real-channel validation deferred to Phase 9 §10.2)
- ☒ Crash-reporting/telemetry SDK wired and consent-gated

**Test Plan (cross-reference):** Integration tests per MVP2\_SECURITY\_PERFORMANCE.md §8.2; security tests (extension permissions, XSS, CSP) per §8.6; relevant success criteria: SC-01, SC-04 (MASQUE — if routed via extension proxy config), SC-20 (UI responsiveness, popup <100ms) per MVP2\_OVERVIEW.md §9.

---

## 10. Phase 9: Integration & Launch (Weeks 32-36)

## 10.1 Phase Overview

Attribute	Value
Duration	5 weeks (overlaps with Phase 8 from Week 32)
Team	All teams (full team effort)
Depends On	At least one client per tier (T1-T4) feature-complete
Blocks	None (final phase)
Goal	Production launch across all 8 platforms

## 10.2 Week-by-Week Breakdown

## Week 32: End-to-End Integration Testing

### Tasks:

- ☐ Cross-platform integration test suite:
    - Automated: Connect → verify IP → run traffic → disconnect → verify IP
    - Platforms: macOS, Linux, Windows, Android, iOS, HarmonyOS, Aurora OS, Web
    - Protocols: WireGuard, Shadowsocks
    - Scenarios: full tunnel, split tunnel, kill switch, DNS leak
  - ☐ Implement integration test harness:
    - Docker-based VPN server (test fixture)
    - `ipleak.net` API for IP verification
    - DNS leak test via multiple resolver checks
    - Speed test via `speedtest-cli` integration
  - ☐ Run full integration matrix:

	Platform	WireGuard	Shadowsocks	Kill Switch	Split Tunnel	DNS	
Test	Test	Test	Test	Test	Test	Test	macOS
Test	Test	Test	Test	Test	Test	Test	Linux
Test	Test	Test	Test	Test	Test	Test	Windows
Test	Test	Test	Test	Test	Test	Test	Android
Test	Test	Test	Test	Test	Test	Test	iOS
Test	Test	Test	Test	Test	Test	Test	HarmonyOS
Test	Test	Test	Test	Test	Test	Test	Aurora OS
Test	Test	Test	Test	Test	Test	Test	Web
  - ☐ Document test results and file tickets for failures
  - ☐

**Production-readiness milestone — MDM/managed-policy validation** (Owner: Rust Core Lead + the relevant platform lead per §11.1; cross-ref MVP2\_ARCHITECTURE.md §10.2). This is the first of the four cross-cutting production-readiness milestones flagged as missing from the pre-hardening roadmap — `apply_managed_policy()` was implemented per-platform in Phases 3-6/8; this week validates it end-to-end against at least one REAL managed- config channel per platform class (not a mock):

- macOS: push a real Configuration Profile via a test MDM (e.g., an Apple Business Manager sandbox or `mdmclient` test profile) → verify `apply_managed_policy()` enforces forced kill switch + allowed protocols
- Windows: push a real Group Policy Object (or Intune test tenant configuration profile) → verify enforcement
- Android: push a real Android Enterprise managed configuration from an EMM console test tenant → verify enforcement
- Chrome: push a real `ExtensionSettings` policy via `chrome://policy` (or a G Suite/Workspace admin test domain) → verify enforcement
- iOS: push a real Apple MDM managed app config (per-app VPN) from a test MDM → verify enforcement
- Exit criteria: each validated channel produces a captured before/after policy-enforcement diff (e.g., kill switch toggled by policy alone, with no local user action) — a channel that cannot be validated end-to-end (no test MDM tenant available) is tracked as `operator-blocked` with the self-resolution-exhaustion trail, never silently marked done

### Deliverables:

- Integration test report (all platforms × all features)
  - Automated test harness
  - Failure tickets created and prioritized
  - MDM/managed-policy validation report (per-platform-class real-channel proof)
- 

## Week 33: Security Audit

### Tasks:



External security audit (3rd party):

- Rust core: memory safety, crypto implementation, FFI boundaries
- Desktop: Tauri IPC security, local storage, update mechanism
- Mobile: secure storage, certificate pinning, obfuscation
- Web: extension permissions, XSS prevention, CSP



Internal security review:

- Dependency audit (cargo-audit, npm audit)
- Secret scanning (no hardcoded keys)
- SAST (static application security testing)
- Code review for all security-critical paths



Penetration testing:

- VPN server interaction (fuzzing, injection)
- Client-side tampering resistance
- Update mechanism integrity verification



Cryptographic review:

- Key generation entropy verification
- Key storage security (platform keystore usage)
- Protocol implementation correctness (WireGuard spec compliance)



Address audit findings:

- Critical: fix before launch
- High: fix before launch or implement mitigations
- Medium: document, fix post-launch
- Low: document for future releases



**Production-readiness milestone — reproducible build + SBOM verification** (Owner: DevOps/CI Engineer; cross-ref MV P2\_SHARED\_CORE.md §5.5, MVP2\_ARCHITECTURE.md §10.3). Phase 1 Week 1 wired the CI-side tooling (cargo build --locked, cargo-cyclonedx/syft); this week re-verifies it against the ACTUAL release-candidate artifacts across all 14+ cross-compiled targets (Appendix A), not just the Phase 1 sample build:

- Rebuild each release-candidate artifact twice from the same tagged commit and diff the binary hashes — reproducibility proof per artifact
- Confirm every artifact ships a generated (never hand-maintained) SBOM attached alongside the signed artifact
- Exit criteria: 100% of release artifacts reproducible + SBOM-attached, or each exception documented with root cause (e.g., a non-deterministic linker step) and a tracked follow-up item — never silently skipped

**Deliverables:**

- Security audit report
  - All critical/high findings addressed
  - Security hardening complete
  - Reproducible-build + SBOM verification report (per release artifact)
-

## Week 34: Performance Optimization

### Tasks:

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Rust core optimization:

- Profile with cargo-flamegraph
- Optimize hot paths (packet encryption/decryption)
- Reduce lock contention in Tunn
- Optimize memory allocations (reuse buffers)

☐

Desktop optimization:

- Bundle size audit and reduction
- Startup time profiling and optimization
- Memory leak detection (72-hour soak)

☐

Mobile optimization:

- Battery drain profiling (Instruments, Android Profiler)
- APK/IPA size optimization
- Cold start time optimization

☐

Network performance:

- Throughput benchmarking (iperf3)
- Latency under load
- Connection establishment time (p50, p95, p99)
- Reconnection time after network change

☐

Performance targets verification:

Metric	Target	Status
Desktop bundle size	< 15 MB	Verify
Desktop startup	< 1.5s	Verify
Desktop idle RAM	< 80 MB	Verify
Mobile bundle size	< 25 MB	Verify
Mobile idle RAM	< 120 MB	Verify
Connection time (p95)	< 3s	Verify
Throughput	> 500 Mbps	Verify
Battery drain (mobile)	< 3%/hour	Verify

## Deliverables:

- Performance benchmark report
  - All targets met or documented exceptions
  - Optimization PRs merged
- 

## Week 35: Documentation Finalization & Beta Release

### Tasks:



Finalize all documentation:

- **User documentation:**
  - Quick start guide (per platform)
  - Connection troubleshooting
  - Kill switch explanation
  - Split tunneling setup
  - FAQ
- **Developer documentation:**
  - Architecture overview (updated)
  - API reference (rustdoc)
  - Contributing guidelines
  - Build instructions (per platform)
- **Admin documentation:**
  - Fleet management guide
  - Configuration deployment
  - Analytics interpretation
- **Ops documentation:**
  - Deployment runbook
  - Monitoring and alerting
  - Incident response procedures



**Production-readiness milestone — consent-gated crash/telemetry pipeline LIVE (blocking gate before beta invites go out)** (Owner: DevOps/CI Engineer + QA Engineer, with Product/Legal sign-off on the consent-copy wording; cross-ref MVP2\_ARCHITECTURE.md §10.1 MONITOR subgraph). Per-platform SDK wiring + consent prompts landed in Phases 3/4/5/6/8; this is the go/no-go check that the pipeline is actually receiving events end-to-end BEFORE any external user is invited:

- Verify Sentry (crash/error) + client API telemetry (connection success/failure, latency) are receiving real events from a test build on every platform
- Verify NO event is emitted from a build where consent was declined (negative test — the consent gate must fail closed)
- Verify the Grafana dashboard (Appendix D) surfaces crash-rate and error-rate deltas per platform — this is the same signal the canary rollback decision (below) depends on



- Exit criteria: pipeline demonstrably live + consent-gate proven both ways (opt-in emits, opt-out is silent) on every platform, captured as evidence



**Production-readiness milestone — staged/canary rollout pipeline rehearsed at least once before GA** (Owner:

DevOps/CI Engineer; cross-ref MVP2\_ARCHITECTURE.md §10.1 for the full canary-ring flowchart — not duplicated here). Use this Beta release itself as the rehearsal:

- Execute the internal ring → canary-1% → canary-10% → canary-50% → GA progression (or the store-native equivalent: Play Console staged rollout, App Store Connect phased release, self-hosted updater ring) against the beta population instead of a synthetic dry run
- Verify the automatic-rollback trigger (crash-rate/error-rate delta breaches threshold → auto-revert to the internal ring) actually fires against an injected synthetic regression in a non-production channel
- Exit criteria: at least one full ring progression completed with the rollback trigger proven to fire on demand; any platform whose distribution channel has no phased-rollout primitive (e.g., a store that only supports all-or-nothing release) documents that constraint explicitly rather than silently skipping the rehearsal



Beta release:

- Desktop: GitHub Releases with signed artifacts
- Mobile: TestFlight (iOS) + Play Store Internal Testing (Android)
- Web: Chrome Web Store beta channel
- Invite beta testers (target: 500 across all platforms)
- Beta feedback collection mechanism (in-app + email)



Monitor beta metrics:

- Daily active users
- Connection success rate
- Crash-free sessions
- Support ticket volume

**Deliverables:**

- Complete documentation set
  - Beta release live
  - Beta monitoring dashboard
  - Consent-gated telemetry pipeline proven live (positive + negative test)
  - Canary/staged-rollout pipeline rehearsed at least once with rollback proven
-

## Week 36: Production Launch

### Tasks:



Final go/no-go decision:

- Review beta metrics (connection success > 95%, crash-free > 99%)
- Confirm all P0 bugs resolved
- Security audit sign-off
- Performance targets met
- Confirm the four Phase 9 production-readiness milestones are GO: MDM validation (Week 32), reproducible-build + SBOM verification (Week 33), consent-gated telemetry pipeline live (Week 35), canary pipeline rehearsed (Week 35) — see §10.2 above and §10.4 Exit Criteria



Production release — executed as a staged/canary rollout per the pipeline rehearsed in Week 35 (internal ring → canary 1% → 10% → 50% → GA 100%, per MVP2\_ARCHITECTURE.md §10.1), monitored against the live crash-rate/error-rate dashboard with auto-rollback armed, not an all-at-once release:

- Desktop: GitHub Releases + auto-update push (staged where the updater supports it; Tauri updater channel ring)
- macOS: Notarized DMG
- Windows: Signed MSI
- Linux: DEB/RPM/Flatpak/AppImage
- Android: Play Store production release (staged rollout percentage via Play Console)
- iOS: App Store release (phased release via App Store Connect)
- HarmonyOS: AppGallery release
- Aurora OS: OpenRepos release
- Web: Chrome Web Store + AMO + Edge Add-ons release
- Final confirmation: every published artifact carries its SBOM + reproducible-build attestation generated in Week 33 (release-signing + provenance, cross-ref MVP2\_ARCHITECTURE.md §10.1 SBOM stage) — a release artifact without an attached SBOM is a launch blocker, not a post-launch follow-up



Launch communications:

- Release notes (per platform)
- Blog post / announcement
- Social media coordination
- Support team briefing



Post-launch monitoring:

- Real-time connection metrics
- Error rate monitoring
- Support queue monitoring

- Social sentiment tracking
- On-call engineering rotation (first 72 hours)

### **Deliverables:**

- All platforms live in production
- Release notes published
- Monitoring active
- On-call rotation scheduled

### **Milestone: M8 — Launch Ready (All Platforms Production)**

## **10.3 Phase 9 Deliverables Summary**

<b>Deliverable</b>	<b>Status</b>
Integration test report	All platforms tested
Security audit	Critical findings resolved
Performance benchmarks	Targets met
Documentation	Complete (user + dev + admin + ops)
Beta release	500 beta testers
Production release	All 8 platforms live
MDM/managed-policy validation	Real-channel proof, macOS/Windows/Android/iOS/Chrome
Reproducible build + SBOM	Verified across all 14+ release artifacts
Consent-gated telemetry pipeline	Live before beta, positive + negative test proven
Staged/canary rollout	Rehearsed on Beta, executed on GA, rollback proven

## **10.4 Phase 9 Exit Criteria**

- ☒ Integration tests passing for all platforms
- ☒ Security audit complete with no critical findings
- ☒ Performance targets verified
- ☒ Documentation complete and published
- ☒ Beta metrics positive (> 95% connection success)
- ☒ All 8 platforms released to production

- Monitoring and on-call active
- **MDM/managed-policy push validated against a real channel per platform class** (macOS Configuration Profile, Windows Group Policy/Intune, Android Enterprise managed configuration, Apple MDM managed app config, Chrome enterprise policy) — Week 32
- **Reproducible build + SBOM generation verified on every release artifact** (14+ cross-compiled targets), attached to every published artifact — Week 33 / Week 36
- **Crash-reporting/telemetry pipeline live and privacy-consent-gated** before any public beta invite, proven both positive (consent → events flow) and negative (no consent → silent) — Week 35
- **Staged/canary rollout pipeline stood up and rehearsed at least once** before GA, with the automatic-rollback trigger proven to fire on an injected synthetic regression, then executed for the actual GA release — Week 35 (rehearsal) / Week 36 (execution)

**Test Plan (cross-reference):** Full-suite integration/E2E per MVP2\_SECURITY\_PERFORMANCE.md §8.2/§8.7 CI test matrix; security/penetration testing per §8.6; relevant success criteria: SC-01 through SC-10 (feature completeness across all 8 platforms), SC-21/SC-22 (security), SC-30 (crash-free rate  $\geq 99.9\%$  over 30-day monitoring) per MVP2\_OVERVIEW.md §9. This phase is where the risk-ordered validation discipline matters most — run the highest-risk, most-recently-touched, most-reopened items first per the project's own change-history, then the remaining suite.

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## 11. Team Structure Recommendations

### 11.1 Recommended Team Composition

Role	Count	Responsibilities	Primary Phases
<b>Rust Core Engineer (Lead)</b>	1	Core architecture, protocols, FFI design, code review	1, 2, all (support)
<b>Rust Core Engineer</b>	1	Platform adapters, crypto, testing, CI/CD	1, 2, all (support)
	1		3

Role	Count	Responsibilities	Primary Phases
<b>Desktop Developer (macOS/Linux)</b>		Tauri app, macOS NE, Linux packaging	
<b>Desktop Developer (Windows)</b>	1	Tauri app, WFP, WinTUN, MSI	4
<b>Mobile Developer (Android)</b>	1	Flutter, VpnService, Play Store	5
<b>Mobile Developer (iOS)</b>	1	Flutter, NEPacketTunnelProvider, App Store	6
<b>HarmonyOS Specialist</b>	1	HarmonyOS + Aurora OS adaptations	7
<b>Web Developer</b>	1	Extension, admin panel, PWA, WASM	8
<b>DevOps/CI Engineer</b>	1	Build pipeline, releases, infrastructure	1, all (support)
<b>QA Engineer</b>	1	Test planning, execution, automation	9 (all phases)
<b>UI/UX Designer</b>	1	Design system, screens, platform adaptations	3, 5, 7, 8

**Total Headcount:** 11 FTE

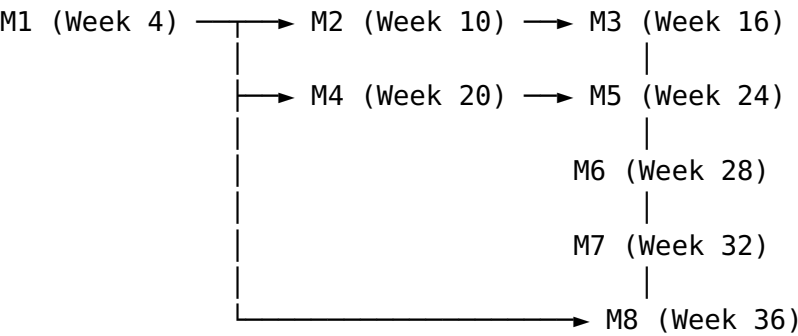
## 11.2 Team Allocation by Phase

Phase	Rust	Desktop	Mobile	HarmonyOS	Web
DevOps   QA   Design					
1. Foundation 1	2   -   -	-	-	-	-
2. Protocols 1	2   -   -	-	-	-	-
3. macOS/Linux 0.5	0.5   0.5   0.5	1	-	-	-
4. Windows 0.5	0.5   0.5   0.5	1	-	-	-
5. Android 0.5	0.5   0.5   0.5	-	1	-	-
6. iOS 0.5	0.5   0.5   0.5	-	1	-	-
7. HarmonyOS 0.5	0.5   0.5   0.5	-	-	1	-
8. Web	0.5	-	-	-	1



Milestone	Date (Week)	Deliverables	Gates
<b>M4: Mobile Alpha</b>		Android client, Play Store ready	AAB < 25MB, VpnService functional
<b>M5: Mobile Beta</b>	Week 24	Android + iOS clients	TestFlight uploaded, both platforms tested
<b>M6: Extended Platforms</b>	Week 28	HarmonyOS + Aurora OS	AppGallery + OpenRepos submitted
<b>M7: Web Complete</b>	Week 32	Extension + Admin + PWA	Chrome Web Store + AMO submitted
<b>M8: Launch Ready</b>	Week 36	All platforms production	Security audit, beta metrics, go/no-go

### 12.2 Milestone Dependency Chain



### 12.3 Milestone Exit Gates

Each milestone has explicit exit criteria that must be met before proceeding:

Gate	Criteria	Verification
<b>Code Quality</b>	No P0/P1 bugs, > 70% test coverage	CI report, bug tracker
<b>Security</b>	cargo-audit clean, no secrets leaked	Automated scan
<b>Performance</b>	Meet phase-specific targets	Benchmark report
<b>Documentation</b>	API docs, user guides updated	Doc review
<b>Sign-off</b>	Tech lead + QA approval	Meeting notes

# 13. Risk-Adjusted Timeline

## 13.1 Scenario Analysis

Scenario	Duration	Probability	Trigger
Best Case	30 weeks	20%	No blocking issues, team performs above expectations
Expected	36 weeks	60%	Normal pace with expected challenges
Worst Case	44 weeks	20%	Major blockers requiring architecture changes

## 13.2 Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation	Contingency
R01	Apple Network entitlement delay	Medium	High	Submit entitlement request in Week 20 (6 weeks buffer)	iOS Beta via TestFlight without NE (limited)
R02	Tauri v2 breaking changes	Medium	Medium	Pin to stable release; maintain fork if needed	Fall back to Tauri v1 or Electron
R03	Flutter HarmonyOS embedding breaking changes	Medium	High	Pin to v3.22.0-ohos; maintain fork	Delay HarmonyOS; use pure ArkTS implementation
R04	Windows WFP conflicts with third-party AV	Medium	Medium	Extensive AV compatibility testing; filter weight tuning	Document known incompatibilities
R05	Rust core binary size exceeds target	Medium	Low	Apply all size optimizations; feature flags for platforms	Accept larger size on affected platforms
R06	WireGuard protocol patent/licensing issues	Low	High	Use boringtun (Cloudflare, permissive license); legal review	Implement Shadowsocks as primary protocol
R07		Medium	High		



Risk ID	Risk Description	Probability	Impact	Mitigation	Contingency
	Key engineer departure			Knowledge documentation; pair programming; bus factor > 1	Contract consultants; adjust timeline
R08	VPN server API changes mid-development	Low	Medium	API versioning; mock server for testing	Adapter layer for API changes
R09	iOS App Store rejection	Medium	High	Clear privacy policy; legitimate use case; comply with entitlements	Appeal process; modify app if needed
R10	Android OEM background kill	High	Medium	Foreground service; battery optimization guides; UIDT	Document workarounds per OEM

### 13.3 Buffer Allocation

Phase	Planned	Buffer	Risk-adjusted
Phase 1	4 weeks	+0	4 weeks
Phase 2	6 weeks	+1	7 weeks
Phase 3	9 weeks	+1	10 weeks
Phase 4	7 weeks	+1	8 weeks
Phase 5	9 weeks	+1	10 weeks
Phase 6	7 weeks	+1	8 weeks
Phase 7	7 weeks	+2	9 weeks
Phase 8	7 weeks	+1	8 weeks
Phase 9	5 weeks	+1	6 weeks
<b>Total</b>		<b>+9</b>	

Phase	Planned	Buffer	Risk-adjusted
	61 work-weeks		70 work-weeks ( $\approx$ 44 calendar weeks with overlap)

**Expected (60% probability):** 36 weeks

**With 2-week management reserve:** 38 weeks

**Worst case (20% probability):** 44 weeks

## 13.4 Early Warning Indicators

Indicator	Green	Yellow	Red
Sprint velocity	> 90% of planned	70-90%	< 70%
Bug backlog	Decreasing	Stable	Increasing
CI failure rate	< 5%	5-15%	> 15%
Code review turnaround	< 24h	24-48h	> 48h
Test coverage	> 75%	60-75%	< 60%
Documentation completeness	> 80%	60-80%	< 60%

## 14. Appendices

### Appendix A: Weekly Sprint Schedule

Monday	Sprint planning (1h) + Team standup (15min)
Tuesday	Development + Mid-week check-in (async)
Wednesday	Development + Code review focus
Thursday	Development + Technical discussion (if needed)
Friday	Demo/retrospective (1h) + Standup (15min)

### Appendix B: Definition of Done

- ☐ Code implemented and self-reviewed
- ☐ Unit tests written and passing (> 70% coverage)
- ☐ Integration tests passing (where applicable)
- ☐ Code review approved by at least 1 senior engineer
- ☐ Documentation updated (rustdoc, README, user guide)
- ☐

- ☐ CI pipeline green (all targets)
- ☐ No cargo-clippy warnings
- ☐ Security scan clean (cargo-audit)
- ☐ QA verification (for user-facing features)

## Appendix C: Communication Plan

Channel	Purpose	Frequency
Daily standup	Blockers, progress, plans	Daily (15min)
Sprint review	Demo, feedback, planning	Weekly (1h)
Architecture review	Technical decisions	Bi-weekly (1h)
Risk review	Risk register update	Monthly (1h)
Executive update	Milestone progress	Bi-weekly (30min)
All-hands demo	Cross-team showcase	Monthly (1h)

## Appendix D: Tooling Stack

Category	Tool	Purpose
Version Control	GitHub	Source code, PRs, issues
CI/CD	GitHub Actions	Build, test, release automation
Project Management	GitHub Projects	Sprint tracking, backlog
Communication	Slack	Team chat, async updates
Documentation	Notion / rustdoc	Technical docs, user guides
Design	Figma	UI/UX design, prototypes
Monitoring	Sentry + Grafana	Error tracking, metrics
Security	cargo-audit, Snyk	Vulnerability scanning

## Appendix E: Glossary

Term	Definition
TUN	

Term	Definition
	Virtual network interface operating at Layer 3 (IP)
<b>WFP</b>	Windows Filtering Platform — kernel-mode packet filtering
<b>NEPacketTunnelProvider</b>	iOS/macOS Network Extension for custom VPN protocols
<b>VpnService</b>	Android API for custom VPN connections
<b>FRB</b>	flutter_rust_bridge — Dart-Rust FFI code generator
<b>UniFFI</b>	Mozilla's multi-language binding generator for Rust
<b>MV3</b>	Manifest V3 — latest browser extension standard
<b>PAC</b>	Proxy Auto-Configuration — JavaScript proxy rules
<b>WASM</b>	WebAssembly — sandboxed binary execution in browsers
<b>Kill Switch</b>	Firewall blocking all non-VPN traffic on disconnect
<b>Split Tunneling</b>	Selective traffic routing through VPN
<b>DoH/DoT</b>	DNS-over-HTTPS / DNS-over-TLS
<b>BFE</b>	Base Filtering Engine — Windows WFP service
<b>ConnMan</b>	Connection Manager — Aurora OS network daemon

*Document compiled: 2026-01-21*

*Based on: MVP2\_ARCHITECTURE.md v1.0,*

*MVP2\_SHARED\_CORE.md v0.2.0*

*Total estimated effort: 42,000 lines of code across 8 platforms*

*Next review: Post-M1 (Week 4) for trajectory adjustment*